Subsidies, Cost Trends, and Pass-Through: Evidence from the Residential Solar Market

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Abstract

Solar subsidies are scheduled to fall as deployment grows, while installation costs change over the same period, so pass-through estimates that rely on subsidy timing can pick up cost trends. I study two Massachusetts programs that pay homeowners per kilowatt-hour, first through tradable certificates and then through a fixed rate. I identify pass-through from differences in the subsidy across roofs that the same installer equips in the same month, which arise because the payment depends on each roof's generation. I find that buyers receive about 85 percent of the subsidy under both designs, whereas a comparison over time suggests a 26-point gap. Installers raise prices only slightly for roofs that earn larger subsidies, which, with log-concave demand, bounds their markups at 2–7 percent of price.

1 Introduction

Residential solar is heavily subsidized. The U.S. federal government alone spent $7.5 billion on solar subsidies in fiscal year 2022, almost all of it through tax credits (U.S. Energy Information Administration, 2023), and states add rebates, net metering, and payments for the electricity a system generates (Borenstein, 2017). Production-based payments are a common design: feed-in tariffs and tradable Solar Renewable Energy Certificates (SRECs) pay owners per kWh, and Massachusetts in 2018 and New Jersey in 2021 replaced market-priced certificates with administratively set payments. Whether any of these subsidies lowers the cost of adoption depends on pass-through, the share of the subsidy that reaches buyers rather than installers. Where installers capture part of it through higher prices, each unit of new capacity costs more and adoption may fall.

Measuring pass-through is difficult because subsidies rarely change while everything else stays fixed. Existing estimates for solar identify pass-through from the timing of the subsidy within one state: step-downs in the California Solar Initiative rebate (Pless and van Benthem, 2019; Dong, Wiser and Rai, 2018), step-downs in Connecticut's rebate (Gillingham and Tsvetanov, 2019), and a change in the federal investment tax credit (Podolefsky, 2013). Solar subsidy schedules, however, are designed to decline as deployment grows. Installation costs also change over time. They tend to fall with experience as hardware becomes cheaper and installers learn, but they also rise and fall with input prices and general inflation. A regression of prices on the subsidy mixes how the subsidy is shared with how costs and the subsidy move together. Estimates of this kind depend on how the path of costs is modeled: in Pless and van Benthem (2019), host-owned pass-through ranges from 0.70 to 1.01 across their specifications of the time trend. Studies of tax incidence in other markets address common cost shocks with a comparison group that the tax does not reach (Kosonen, 2015; Kopczuk et al., 2016; Butters, Sacks and Seo, 2022), but a solar subsidy reaches every installation in a utility territory at a given time.

This paper instead compares buyers who receive different subsidies in the same market at

the same time. I study two production-based programs in Massachusetts. Under the Solar Carve-Out II program (2014–2018), owners earned SRECs sold at market prices. Under its successor, the Solar Massachusetts Renewable Target (SMART) program, they receive a fixed rate per kWh that is set when the application is approved and declines as capacity blocks fill. Because both programs pay per kWh, a project's subsidy depends on how much electricity its roof will generate. I merge regulatory data from the Massachusetts Department of Energy Resources (DOER) with Lawrence Berkeley National Laboratory's Tracking the Sun database for more than 39,000 host-owned residential systems installed between 2014 and 2023, geocode each project's street address, and model each system's annual generation with the National Renewable Energy Laboratory's PVWatts calculator from its location, size, tilt, azimuth, and equipment. About 91% of the variance in modeled capacity factors is within towns. Roofs that the same installer equips in the same month therefore earn different subsidies, and this variation does not depend on the policy calendar. My preferred specification compares roofs within installer-by-year, town-by-year, and month cells. Comparing roofs of the same installer in the same month also gives the same pass-through results. This design is available because the subsidy is paid for production, while an upfront rebate per watt is nearly the same for every roof.

The first finding is that the two designs pass through the subsidy at the same rate. Comparing roofs, pass-through is 0.86 under Solar Carve-Out and 0.85 under SMART. The difference is −0.02 with a standard error of 0.085, and the 95% confidence interval runs from −0.18 to 0.15. The subsidy in both programs is built from the same model of generation, so error in modeled generation does not by itself create or hide a difference between them. It does pull both levels toward one, so true pass-through may be somewhat lower. Installers' own production estimates, which SMART applications report, agree only partly with modeled generation within these cells, and using them as an alternative measure gives SMART pass-through of 0.74–0.77. Those estimates are prepared for sales and exist only for SMART, so I treat them as a sensitivity check. The programs differ in the form of the payment, in

how predictable it is, and in size: the average Solar Carve-Out subsidy is about 3.4 times the average SMART subsidy per kW ($1,923 versus $566 in present value), about 45% and 15% of the price. Buyers nonetheless capture about 85 cents of each dollar under both. To my knowledge, these are the first estimates of the pass-through of a production-based subsidy and the first comparison of a certificate-based and a fixed-rate design within one market.[1]

The second finding is that estimates that rely on the timing of the subsidy follow the cost trend. Under SMART, the subsidy kept falling after 2020 while prices rose, which biases a timing-based estimate up. A naive comparison with year fixed effects gives 1.02. A stacked event study around the step-downs in SMART rates designed to avoid trends, gives 1.27. Both are against the across roofs comparison. Measurement error cannot produce this gap, because it pulls estimates toward one rather than above it, and pass-through above one is inconsistent with the log-concave demand I estimate unless marginal cost falls with volume. Under Solar Carve-Out, the subsidy and installation prices fell together, which biases a timing-based estimate down. The naive estimate, 0.76, lies below the cross-roof estimate of 0.86, as predicted, but error in modeled generation also pulls the cross-roof estimate toward one, and the estimate identified from roof orientation alone is 0.76. The data therefore cannot show that the naive Solar Carve-Out estimate is biased, but the timing-based estimates move with the cost trend in their window: a regression discontinuity at the January 2017 cut in the SREC factor falls from 1.03 to 0.71 as the window widens from 90 to 365 days. The naive gap of 26 percentage points between the programs therefore comes mostly from the SMART estimate, and year-month fixed effects alone remove most of it.

These results also bound one source of installer market power: price discrimination across roofs according to the subsidy each roof earns. Under a uniform price, a subsidy specific to one roof would be passed through in full, so the shortfall of cross-roof pass-through below one measures how far installers price the roof. Installers quote each job individually, and

[1] Comparisons in other settings find that the form of an incentive can affect pass-through: car buyers capture 70 to 90 percent of a customer rebate but 30 to 40 percent of a dealer discount (Busse, Silva-Risso and Zettelmeyer, 2006), and the point at which a diesel tax is collected changes its pass-through (Kopczuk et al., 2016).

prices differ by about $450–490 per kW across buyers of the same installer in the same month, yet a roof that earns $59–137 more modeled subsidy per kW is charged only $6–20 more. With log-concave demand under SMART, the sufficient-statistics framework of Weyl and Fabinger (2013) turns this into a bound: installer markups are at most 2–7% of price, and they exceed 10% only if more than about half of the remaining variation in modeled generation is error. The bound rests on demand curvature that is estimated with limited precision. The other channels through which installers could earn markups, such as local concentration, the number of competitors, and buyer income, leave little trace in pass-through or prices, although SMART installers price the roof somewhat more in months of high statewide demand. Roof-specific pass-through cannot detect a markup that installers charge every buyer alike, regardless of the roof's subsidy, such as one sustained by limited competition or spent on acquiring customers: such a markup leaves pass-through at one, so the bound applies to price discrimination, not to the overall level of markups.

This paper contributes to three literatures. The first studies the incidence of taxes and subsidies and the variation that identifies it. Reduced-form studies identify incidence from changes in taxes and subsidies over time, from rebate step-downs and tax credit changes (Sallee, 2011; Busse, Silva-Risso and Zettelmeyer, 2006) to tax changes compared with untaxed goods or regions (Kosonen, 2015; Kopczuk et al., 2016; Butters, Sacks and Seo, 2022), from national credit prices (Lade and Bushnell, 2019), and from before–after comparisons (Kenkel, 2005). The variation also determines the what pass-through measures: in U.S. oil refining, pass-through ranges from near zero for a cost shock specific to one firm to about one for a shock common to the industry (Muehlegger and Sweeney, 2022), and California's electric vehicle rebates, which differ by income, identify the price recipients pay relative to non-recipients (Muehlegger, Rapson et al., 2018).[2] I show that in a market where the sub-

[2]Related work studies hybrid vehicle subsidies with endogenous quality and costly bargaining (Gulati, McAusland and Sallee, 2017), electric vehicle subsidies across countries (Barwick et al., 2023), the incidence of agricultural subsidies on farmland rents (Kirwan, 2009), and the welfare effect of a consumer subsidy restricted by price ceilings (Fan and Zhang, 2022). Incomplete pass-through can also reflect local costs and markup adjustment (Nakamura and Zerom, 2010), and firms that set prices nationally may respond differently to local cost shocks (Butters, Sacks and Seo, 2022).

sidy schedule tracks costs and no untreated group exists, comparing buyers who face different subsidies at the same time recovers pass-through, and that timing-based designs, including a regression discontinuity and an event study, move with the cost trend in their window. Where prices rose as the subsidy fell, they give pass-through above one.

The second literature estimates the pass-through of solar subsidies and studies how subsidy design affects buyers. Most of this evidence comes from upfront rebates, and it finds that buyers capture most but not all of the subsidy: 78% for host-owned systems in California (Pless and van Benthem, 2019), 84% in Connecticut (Gillingham and Tsvetanov, 2019), and close to complete pass-through in other work on California (Dong, Wiser and Rai, 2018). Pless and van Benthem (2019) find more-than-complete pass-through for third-party-owned systems, and for the federal investment tax credit, reported prices also reflect the incentive to overstate system costs (Podolefsky, 2013). Solar adoption responds strongly to rebates (Hughes and Podolefsky, 2015; Crago and Chernyakhovskiy, 2017), buyers discount production-based payments heavily (De Groote and Verboven, 2019), and upfront subsidies induce more adoption while production subsidies are more efficient (Burr, 2016).[3] I provide estimates for production-based subsidies paid through certificates and through fixed rates, and I find that the two designs pass through the same share once the timing of the subsidy is absorbed.

The third literature uses pass-through to learn about market power. Pass-through tends to rise with competition, from 13% in the least competitive Medicare Advantage markets to 74% in the most competitive (Cabral, Geruso and Mahoney, 2018) and, for fuel taxes on Greek islands, from 0.4 in monopoly markets to 1 in markets with four or more competitors (Genakos and Pagliero, 2022). It has been used to measure incidence under imperfect competition (Ganapati, Shapiro and Walker, 2020) and, with convex demand, to detect mar-

[3]Other studies compare the effectiveness of residential solar incentives (Matisoff and Johnson, 2017), estimate how residential solar demand depends on net metering and electricity prices (Ros and Sai, 2023), and characterize optimal subsidies for residential solar (Colas and Reynier, 2026). Consumer subsidies can also lower production costs through induced innovation (Gerarden, 2023), and the phase-out of a subsidy shapes the dynamics of demand (Hu, Yin and Zhao, 2023).

ket power from more-than-complete pass-through (Pless and van Benthem, 2019), while high pass-through can also arise from cost shocks common to all firms (Fabra and Reguant, 2014). I bound markups with roof-specific pass-through, which removes the supply side from the pass-through formula, together with estimated log-concave demand.

Section 2 describes the two programs, Section 3 the data and the construction of the subsidy, and Section 4 the empirical strategy and why timing variation is confounded. Section 5 compares pass-through across the two programs, Section 6 shows that the estimates that rely on timing follow the cost trends, and Section 7 addresses threats to the cross-roof design. Section 8 draws the implications for market power, and Section 9 concludes.

2 Institutional Background

When a household in Massachusetts installs a solar system, one of two production-based subsidies is available, depending on the policy regime at the time of installation. If a qualifying system entered commercial operation between January 2008 and November 2018, the owner was eligible to receive certificates under the Solar Carve-Out program, which launched in 2010 and covered systems operating since 2008. If it entered commercial operation after November 2018, the owner is entitled to an incentive payment through the electricity bill, on top of net metering, under the SMART program. This section describes the two programs, and Section 3 describes the data and the method I use to compute the subsidy.

2.1 Solar Carve-Out

The Solar Carve-Out is part of Massachusetts's Renewable Portfolio Standard (RPS) and requires that a certain percentage of electricity come from solar resources. The program launched on January 1, 2010, and its requirement increased over time until it reached a capacity target of 400 MW. The state expanded the program in April 2014 with Solar Carve-Out II, which set a total goal of 1,600 MW of solar PV capacity.

Energy suppliers, including utilities and competitive energy suppliers, are required to purchase Solar Renewable Energy Certificates (SRECs) to demonstrate compliance or make an alternative compliance payment (ACP). The NEPOOL Generation Information System (NEPOOL GIS) tracks solar generation and issues SRECs quarterly to solar generation unit owners at a rate of 0.45–1.0 SREC per MWh, depending on system characteristics (the SREC factor). For example, a system smaller than 25 kW authorized to interconnect before January 8, 2017, receives a factor of 1.0. Each qualified solar generation unit is entitled to receive SRECs for 40 calendar quarters as long as the unit remains operational. After 40 calendar quarters, the system earns Renewable Energy Certificates (RECs) instead, which are much less valuable than SRECs.[4][5]

Solar generation unit owners can either market and trade SRECs independently or join an aggregator that harvests and trades SRECs on their behalf. Trading platforms and aggregators typically charge fees of 1–10% of the proceeds depending on system size, with small residential systems (under 50 kW) commonly paying 10%. The Massachusetts Department of Energy Resources (DOER) holds an annual fixed-price clearinghouse auction from May 16 to June 15 for SRECs not sold during the previous year. The auction does not guarantee a sale: certificates that remain unsold through it can still be traded in the market for the next two compliance years. The ACP thus caps SREC prices, but the auction price is not a floor. In the Solar Carve-Out's launch year, the ACP was set at \$600/MWh, while the net clearinghouse auction price was \$285/MWh (a \$300 price less the 5% auction fee).[6]

For Solar Carve Out II, The ACP and auction price schedules were published at launch, whereas under the original program the ACP was adjusted over time. Figure 1 shows historical market prices for SREC II certificates alongside the ACP and clearinghouse auction

[4]The vintage of a certificate is the calendar year in which the electricity is generated. For instance, electricity generated during 2014 earns MA2014(II) certificates, where "MA" denotes Massachusetts, "2014" the generation year, and "II" the program phase, Solar Carve-Out II. Certificates of different vintages form distinct market segments and can trade at different prices, depending on factors such as program caps, ACP levels, and the balance between supply and demand within each vintage year.

[5]An extreme example: in a state that RPS is not binding, such as Texas, RECs have close to zero value.

[6]Under the original program, market prices for SRECs tracked closer to the ACP, and returns exceeded twice the retail electricity price in 2010.

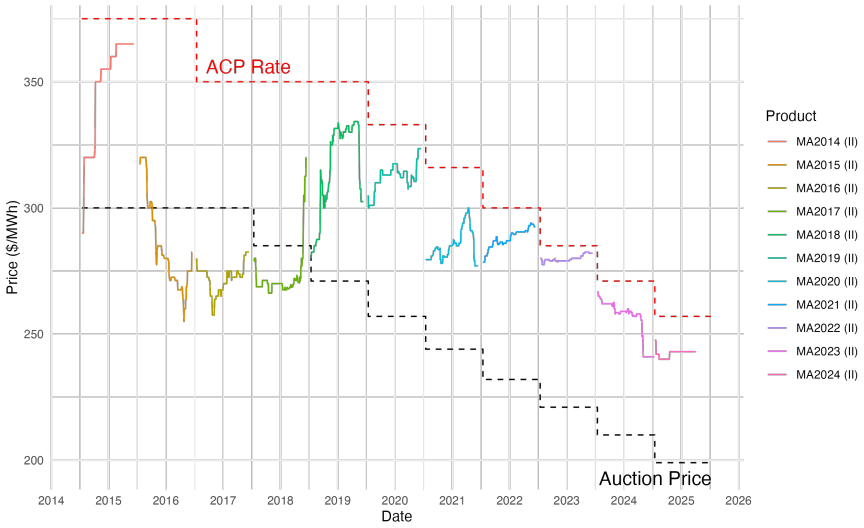


Figure 1: SREC II Market Prices with ACP and Auction Prices. The auction price is plotted gross of the 5% auction fee. An owner selling through the auction receives 95% of the plotted value ($285/MWh in 2014–2016). Each vintage is shown over its trading window, from July 15 of the vintage year to July 14 of the following year.

prices. Publishing the schedules narrowed the range of likely SREC prices but did not remove price risk. In the first few years, market prices were consistently below auction prices because of oversupply. In later years, market prices for SRECs followed the declining trend of the ACP. The net clearinghouse price remained constant at $285/MWh from 2014 to 2016 and then fell by 5% per year.[7]

2.2 SMART

DOER created the Solar Massachusetts Renewable Target (SMART) program to further encourage solar development in Massachusetts after the Solar Carve-Out II target was met. The program launched on November 26, 2018. Under SMART, the three investor-owned utilities (IOUs) in Massachusetts (Eversource, National Grid, and Unitil) compensate solar system owners for the electricity their systems generate, in exchange for the RECs attached to this generation.

The actual incentive rate for a given system is determined by four factors: the base com-

[7]The net clearinghouse price stood at $189.05/MWh in 2024.

pensation rate, household income status, whether the system includes a storage component, and the value of energy. The base compensation rate starts high and declines on a predetermined schedule as capacity blocks fill, and both block sizes and rate schedules vary by utility. The value of energy is set from three-year average retail electricity prices (including delivery and transmission charges) at the time of application. The SMART incentive rate equals the base compensation rate minus the value of energy. Small residential solar generation unit owners receive payments through the utility equal to this rate multiplied by their monthly generation, in addition to the bill savings their behind-the-meter systems already provide, and they can cash out a negative balance monthly. As electricity prices rise and capacity blocks advance, the SMART incentive rate decreases until it reaches zero.[8]

Table 1 shows the base compensation rates for generation units of 25 kW AC or less, by utility.[9]

Table 1: SMART Base Compensation Rates for Generation Units of 25 kW AC or Less

Utility	Low income	Block 1	2	3	4	5	6	7	8
Fitchburg	Yes	\$0.358	\$0.326	\$0.298	\$0.272	—	—	—	—
	No	\$0.311	\$0.284	\$0.259	\$0.236	—	—	—	—
Mass Electric	Yes	\$0.358	\$0.344	\$0.330	\$0.317	\$0.304	\$0.292	\$0.280	\$0.269
	No	\$0.311	\$0.299	\$0.287	\$0.275	\$0.264	\$0.254	\$0.244	\$0.234
Nantucket	Yes	\$0.391	\$0.328	—	—	—	—	—	—
	No	\$0.340	\$0.286	—	—	—	—	—	—
Eversource E.	Yes	\$0.391	\$0.375	\$0.360	\$0.346	\$0.332	\$0.319	\$0.306	\$0.294
	No	\$0.340	\$0.326	\$0.313	\$0.301	\$0.289	\$0.277	\$0.266	\$0.255
Eversource W.	Yes	\$0.329	\$0.315	\$0.303	\$0.291	\$0.279	\$0.268	\$0.257	\$0.247
	No	\$0.286	\$0.274	\$0.263	\$0.253	\$0.243	\$0.233	\$0.224	\$0.215

Base compensation rates in dollars per kWh by utility and capacity block. "Low income" indicates the rates for low-income customers.

SMART was most favorable early in its life cycle, when the incentive rate was considerably higher than the price of RECs. In my sample, the SMART incentive rate is at most

[8]System owners who do not wish to enroll in SMART can instead participate directly in the REC market, and DOER advises them to do so when the SMART rate becomes low. I do not observe how many owners opt out, because owners who do so do not appear in the SMART regulatory data.

[9]There are eight blocks for the three large utilities but only two and four for Nantucket and Fitchburg, respectively. Fitchburg is Unitil's operating company, and Mass Electric and Nantucket Electric are National Grid's operating companies in Massachusetts.

$0.20/kWh, with a median of about $0.066/kWh, while RECs traded at around $0.04/kWh.[10] SMART also offers more predictable cash flows than the Solar Carve-Out program, because the rate is fixed and does not fluctuate with market conditions. The subsidy can therefore be calculated precisely from modeled generation. Another key difference between SMART and Solar Carve-Out is that SMART requires system interconnection to one of the three IOUs. As a result, households served by municipal utilities are ineligible for the program.

2.3 Comparison of the Two Programs

Table 2 summarizes the two programs. Four differences could affect how subsidy value is divided between buyers and installers. First, the form of payment: an owner under Solar Carve-Out sells certificates whose price is set in a market capped by the ACP, whereas a SMART owner receives a fixed per-kWh rate. Second, the timing of information: the SMART rate is fixed when the application is approved, so the buyer and installer know the subsidy per kWh for ten years, while the SREC price the owner will eventually receive is unknown at the time of purchase. Third, the size of the subsidy: the mean present value is about $1,923 per kW under Solar Carve-Out and $566 per kW under SMART, roughly 45% and 15% of the average gross price, with almost no overlap between the two distributions. Fourth, coverage: Solar Carve-Out is open to all utility territories, while SMART is limited to customers of the three IOUs.

These differences suggest that pass-through could differ between the programs. A larger subsidy that arrives through a volatile market price is harder to price into a quote and may be more likely to strain installers' capacity, which would lower pass-through. A small, certain subsidy is easy to incorporate into a price and is less likely to move installer capacity, which would raise pass-through. Because the programs also differ in years, technology costs, and eligible households, a comparison of the programs over time cannot test these predictions. I later tests whether pass-through differs once the timing of the subsidy is absorbed in

[10] Source: SRECTrade trading platform.

Table 2: Comparison of the Solar Carve-Out II and SMART Programs

	Solar Carve-Out II	SMART
Period in sample	Jan. 2014–Nov. 2018	Mar. 2019–Sep. 2023
Form of payment	Tradable SRECs	Fixed rate per kWh
Rate	Market price, capped by the ACP	Fixed at approval
Payment term	40 calendar quarters	10 years
Payer	Retail electricity suppliers	The three IOUs
Eligibility	All utility territories	IOU customers only
Mean subsidy (NPV, 8%)	$1,923 per kW	$566 per kW
Share of gross price	~45%	~15%

Subsidy is the present value per kW of DC capacity at an 8% discount rate with 0.5% annual degradation, from the analysis samples. Shares are ratios of the sample mean subsidy to the sample mean gross price per kW (systems of 25 kW or less).

Section 5.2.

Two rules within the programs change the subsidy discretely, and I use them both in Section 6. On January 8, 2017, the SREC factor for new small systems fell from 1.0 to 0.8, so a system that generates the same electricity receives 20% fewer certificates. Under SMART, the base rate of a utility steps down when one of its capacity blocks fills as in Table 1.

3 Data

3.1 Data Sources and Sample

I obtain the solar installation data from DOER. For SMART projects, the DOER data report compensation rates as well as project characteristics, including location, utility, size, installation cost, installer, and technical details. The technical details, which include azimuth, tilt, and inverter rating, are essential for modeling generation accurately. The compensation rate reported in the data is the actual SMART incentive rate that system owners receive, which reflects any adders and the value of energy.

For Solar Carve-Out II projects, DOER reports similar variables but without technical details. To model annual generation as accurately as for SMART, I merge the DOER data with the Tracking the Sun database published by Lawrence Berkeley National Laboratory (LBNL).[11] All projects in my data appear in Tracking the Sun, along with their technical details. I match projects across the two sources on operation date, size, and total cost, which gives an almost one-to-one match (Appendix A). SREC prices are not available in the project data. I construct an SREC price series from a major SREC trading platform, SRECTrade, and from the price support regulations published on DOER's website.

The SREC sample spans January 1, 2014, to November 26, 2018, the period when Solar Carve-Out II was open to applications. The SMART sample begins on March 25, 2019, marking the debut of the incentive program, and ends on September 20, 2023, by which time most projects received zero compensation and the number of applications had dropped sharply. For SMART projects, I date each project by its Commercial Operation Date (Appendix A).

The sample includes only host-owned, small-scale (≤25 kW DC) residential projects without battery storage, for three reasons. First, the general net metering program has a capacity cap, so projects installed after the cap was reached might not be eligible for net metering. Small systems are exempt from the cap, so restricting the sample to them ensures that every system is on net metering. Second, only host-owned projects have directly observed prices. For third-party-owned projects, the price reported to DOER is reportedly a "market" value rather than the true cost to system owners (Pless and van Benthem, 2019). Third, installations with battery storage qualify for a storage adder, which raises the incentive, and the storage components would distort the measured cost of the solar system. I also remove observations with a per-kW price above the 99th percentile or below the 1st percentile to exclude outliers.[12] Figure 2 shows the location of each project. The main difference between the two

[11] Tracking the Sun is a cleaned dataset that covers more than 60% of solar projects in the United States. LBNL cleans and consolidates information on solar projects from state energy agencies annually. DOER and the Massachusetts Clean Energy Center are both major data providers for LBNL.

[12] Winsorizing the price per kW at the 1st and 99th percentiles instead of dropping these projects leaves the estimates essentially unchanged.

samples is coverage: households in municipal utility territories are ineligible for SMART.[13] In both samples, the density of projects closely resembles the population distribution of Massachusetts.

3.2 Modeling Annual Generation

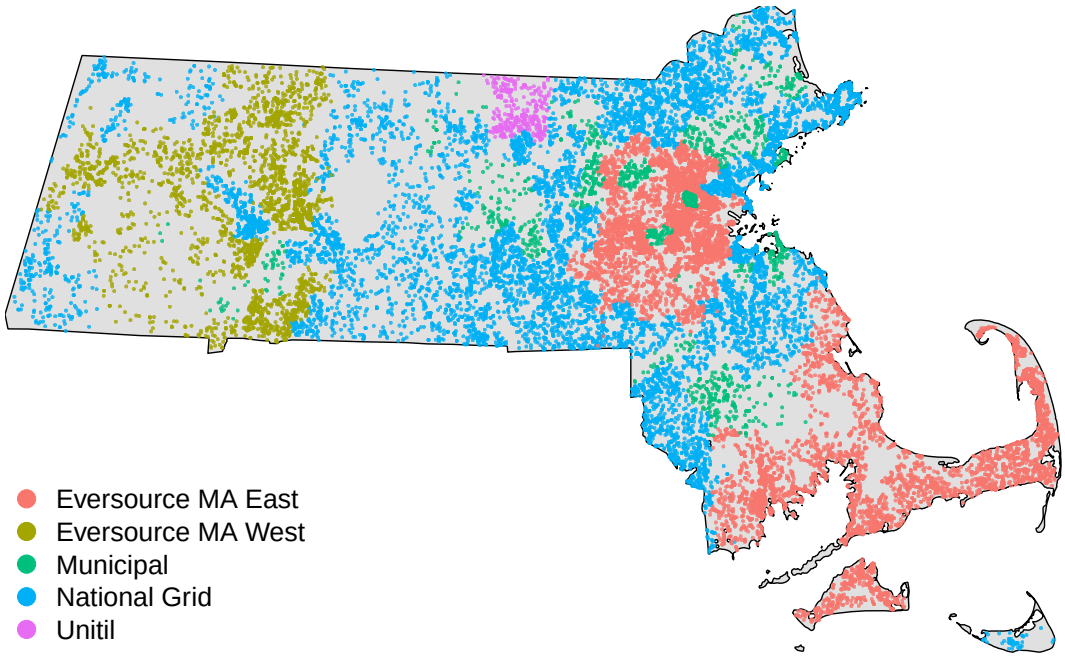
Measuring pass-through requires the value of the subsidy for each project, which the data do not report. Because both the SMART and Solar Carve-Out II programs are production-based subsidies, the total subsidy depends on how much electricity a system will generate and when the payments are made. SMART payments are made monthly through the utility, so the subsidy is the incentive rate times monthly generation. Under the SREC program, payment comes from selling certificates, which are issued every three months, so the subsidy is the market price of certificates times the number of certificates produced each quarter. For simplicity, I calculate the subsidy on an annual basis, which requires annual generation for each project.

I model annual generation using the PVWatts calculator, a tool developed by the National Renewable Energy Laboratory (renamed the National Laboratory of the Rockies in December 2025) for modeling the energy production of grid-connected photovoltaic systems. The calculator combines solar irradiance and meteorological data for a project's location with a set of system parameters to compute monthly generation.

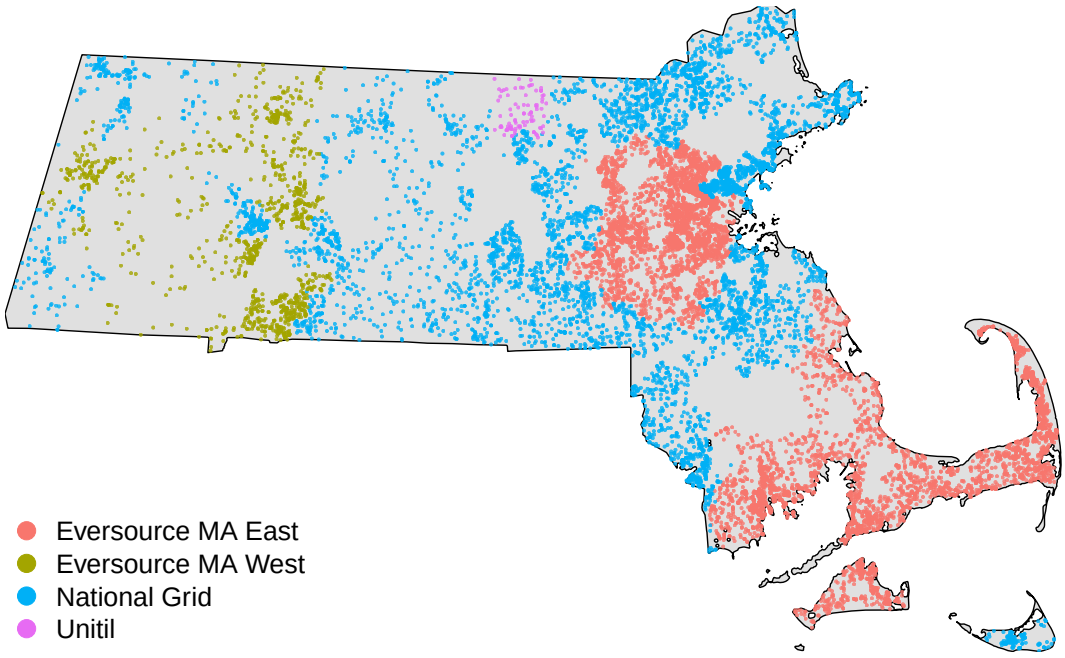
The most important inputs for the calculator are a project's size and location, which determine the area of solar panels and the available solar resource. The public DOER data contain only ZIP codes, which can cover large areas relative to the resolution of the solar resource data, so I obtained each project's street address from DOER and converted it to coordinates (Appendix A).

Besides size and location, the calculator requires the DC-to-AC ratio, module type, array type, azimuth, and tilt. I use the observed values where available, assume a fixed roof-mount

[13] Excluding the 1,567 Solar Carve-Out projects in municipal utility territories changes the Solar Carve-Out estimates very little.



(a) Solar Carve-Out II



(b) SMART

Figure 2: Project Locations by Utility Territory. Each dot is one project in the analysis sample. Municipal utility customers are ineligible for SMART.

array with standard modules, and keep the system loss parameter at the calculator default of 14.08%.[14] The calculator does not model site-specific losses from shading, snow cover, soiling, and outages beyond these defaults, so its output is a model prediction, not a measurement of what each system generates.

The output includes monthly and annual generation for each project. I summarize it using the capacity factor: a system's annual generation as a share of what it would produce if it ran at full rated capacity all year. Most projects have modeled capacity factors of 12.5–15%, and a few are much lower.[15]

What matters most for this study is how generation differs across roofs, and Figure 3 shows that these differences are large. The modeled capacity factor has a standard deviation of 0.012 around a mean of 0.143: more than half of projects (53%) differ from the statewide mean by more than 5%, and 18% by more than 10%. These differences are local. About 91% of the variance in the capacity factor is within towns, so it comes mostly from the orientation, tilt, and design of individual roofs rather than from regional differences in sunshine. Modeling each roof from its own address and design, rather than from its town or ZIP code, is what makes the comparison across roofs possible.

3.3 Subsidy Calculation

I first calculate the net present value of the subsidy for SMART projects, assuming a discount rate of 8%. Because the SMART compensation rate is locked in for ten years when the application is approved, the owner's compensation in a given year depends only on that year's generation. De Groote and Verboven (2019) find that system owners heavily discount the future cash flows from such subsidy programs, so I also report results for other discount

[14] I enter the coordinates as the location input. System size is available for every project, but other parameters are missing for some projects. For a missing parameter, I calculate an implied value. For example, I compute the DC-to-AC ratio for SMART projects from their DC and AC ratings. When no implied value is available, I use the calculator's default values, which represent typical rooftop systems. Fewer than 100 projects require default values.

[15] By entering the technical specifications into the calculator manually, I verified that the low values are correct. They come from suboptimal configurations: a north-facing system, for example, generates much less than a south-facing one, all else equal.

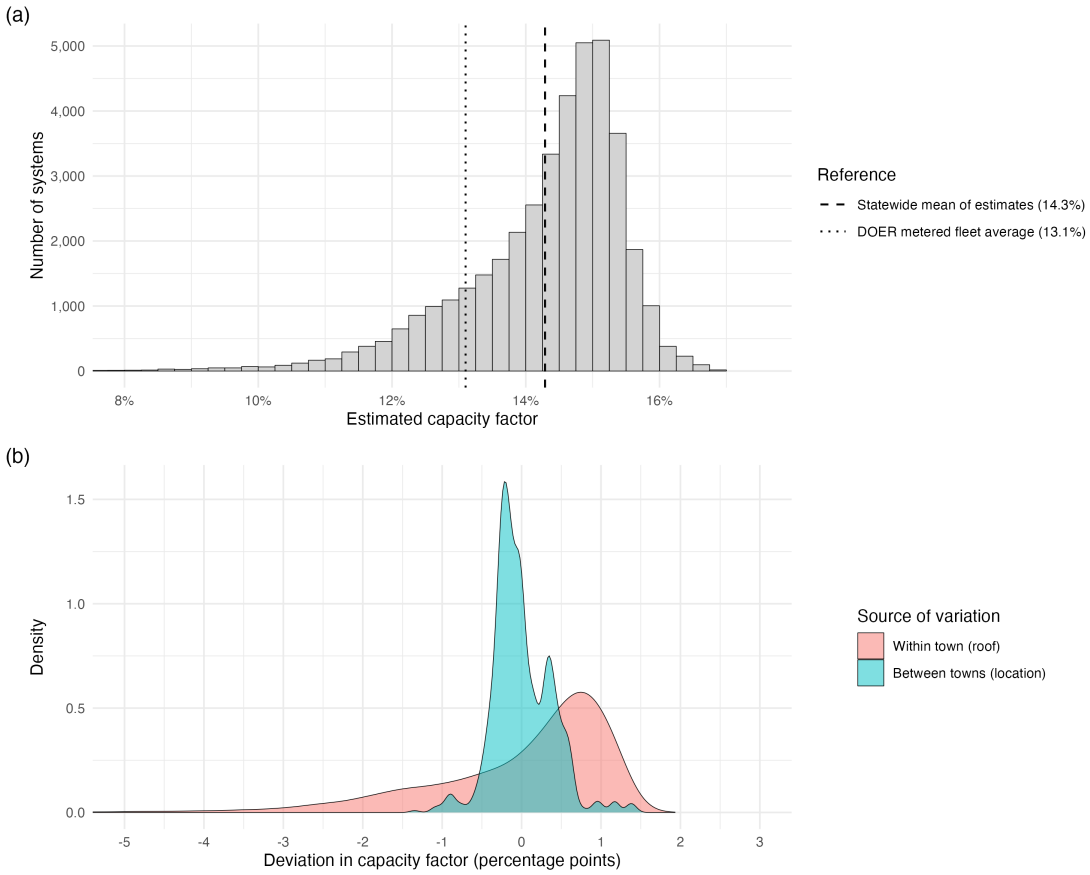


Figure 3: Variation in Modeled Capacity Factor Across Projects

rates. I follow a standard discounted cash flow model:

$$\text{NPV}_i = \sum_{t=1}^{10} \frac{r_i \times g_i (1-a)^{t-1}}{(1+d)^t}$$

where r_i is the SMART incentive rate, g_i is modeled annual generation, d is the discount rate, and a is the panel degradation rate. I assume annual degradation of 0.5% to account for the performance loss from panel aging. Because r_i and g_i do not change over the ten years, the formula reduces to $\text{NPV}_i = r_i \, g_i \, A(d)$, where $A(d) = \sum_{t=1}^{10} (1-a)^{t-1}/(1+d)^t$ is the same for every project. The discount rate therefore rescales every SMART subsidy by the same factor, which is relevant for interpreting the robustness checks in Appendix B.1.

For Solar Carve-Out projects, I apply the same approach with certificate prices:

$$\text{NPV}_i = \sum_{t=1}^{10} \frac{g_i \times f_i \times 90\% \times P_t(1-a)^{t-1}}{(1+d)^t}$$

where g_i is modeled annual generation in MWh, f_i is the project's SREC factor, P_t is the SREC price in year t, and the 90% term reflects the 10% aggregator fee. Ten years of payments cover the whole entitlement of 40 calendar quarters.

Because SREC prices were set in a market, they are not in the Solar Carve-Out data, and the price an owner would receive was not known at the time of purchase. In the early years, market prices were well below the ACP and even below the auction price, so using either alone could misstate the subsidy. My main price path is the midpoint of the ACP and the auction price in each year, the price a buyer would expect without a view on where the market would settle between them. As an alternative, I use the annual average market price, which is what owners actually received, with the 2024 average for years not yet observed.[16] Appendix B.1 compares these and other price paths.

3.4 Validating Modeled Generation

The average modeled capacity factor is 14.3%, slightly above the 13.1% that DOER reports from metered generation for the whole Massachusetts fleet in 2010–2023.[17] A difference of this kind, common to all projects, changes the scale of the subsidy rather than its variation across roofs, and Section 4.5 shows that it moves the pass-through estimates very little.

What matters more is whether modeled generation ranks roofs correctly, and a second

[16] Projects that began operating after 2015 are still receiving SRECs, but their future prices are not yet known. For example, a 2018 project will keep receiving and selling SRECs until 2028, but SREC prices for 2026 through 2028 are still unknown.

[17] Panel degradation explains only part of this gap. Field evidence suggests that panels lose about 0.5% of output per year (Deline et al., 2024), and applying that rate over the age distribution of the fleet closes roughly a fifth of the difference. The larger part of the gap most plausibly reflects losses that PVWatts does not model. The default snow loss of zero is unrealistic for Massachusetts, and the default shading and soiling losses of 3% and 2% are likely too low given the long pollen season and restrictions on tree removal. The statewide fleet also includes lower-quality installations that predate my sample.

measure of generation makes this possible to check. Each SMART application reports an estimated annual production for the system, which the applicant, usually the installer, computes with its own design tools; I call it the installer's production estimate. The estimate is reported for every SMART project. I drop the 1% of projects whose estimate is less than half or more than one and a half times the PVWatts value, which are clear reporting errors such as an annual production of 6 kWh for a 15 kW system. For the remaining projects, the median estimate is 6% below the PVWatts value. Across all projects, the subsidies per kW computed from the two measures have a correlation of 0.97, mainly because both use the SMART rate. Among the roofs the analysis compares, after removing differences across installers, towns, and months (Section 4.4), the two capacity factors have a correlation of only 0.57. If the errors in the two measures are independent, the share of this variation in the modeled capacity factor that is true variation is 0.57, and for the modeled subsidy, which also varies with the SMART rate, it is 0.80. If the errors are positively correlated, for example because both measures use the same recorded tilt and azimuth, these shares are upper bounds. The installer's estimate is not a cleaner measure, however. It is prepared for the sale, it is computed with tools and assumptions that differ across installers, and its error can be correlated with the price: an installer that overstates output may also charge more. It is also unavailable for Solar Carve-Out. The analysis therefore measures generation with PVWatts in both programs, so that the subsidy is built the same way for every roof, and Section 7.2 uses the installer's estimate only as a sensitivity check.

3.5 Summary Statistics

Table 3 presents summary statistics for installations under the Solar Carve-Out program (Panel A) and the SMART program (Panel B). The largest difference between the programs is in the subsidy per kW. Solar Carve-Out installations received an average subsidy of $1,923 per kW, about 3.4 times the mean SMART subsidy of $566 per kW. As a share of the average gross price per kW, the subsidy is about 45% under Solar Carve-Out and 15% under SMART.

This gap reflects the design of the two programs: Solar Carve-Out certificates were priced by a market that tracked a high ACP, whereas SMART rates fall with the value of energy and as capacity blocks fill.

Table 3: Summary Statistics for SMART and Solar Carve-Out Samples

Variable	N	Mean	Std. Dev.	Min	Max
Panel A: Solar Carve-Out Sample					
Size (kW DC)	25,491	7.96	3.20	1.10	25.00
Total Price ($)	25,491	33,376.11	13,130.15	5,175.00	138,300.00
Price per kW ($/kW)	25,491	4,303.56	963.92	2,608.75	7,847.21
Total Subsidy ($)	25,491	15,169.04	6,435.32	1,471.30	60,419.11
Subsidy per kW ($/kW)	25,491	1,922.77	336.09	757.12	2,631.60
Net Price ($)	25,491	18,207.07	8,561.03	796.19	94,096.40
Net Price per kW ($/kW)	25,491	2,380.79	915.74	183.45	6,300.17
SREC Factor	25,491	0.92	0.10	0.80	1.00
Tilt	25,491	29.39	9.13	0.00	89.00
Azimuth	25,491	181.01	41.31	5.00	356.00
N Inverter	25,491	11.86	14.37	1	280
Annual Generation	25,491	10,062.57	4,105.70	1,150.13	33,998.16
Capacity Factor	25,491	0.14	0.01	0.05	0.17
Panel B: SMART Sample					
AC (kW)	14,317	7.49	2.64	1.20	23.50
Size (kW DC)	14,317	9.22	3.53	1.28	24.96
Total Price ($)	14,317	34,305.30	14,175.21	5,517.00	122,655.00
Price per kW ($/kW)	14,317	3,795.74	881.11	1,998.32	7,008.21
Total Subsidy ($)	14,317	5,182.54	3,705.16	0.00	31,917.29
Subsidy per kW ($/kW)	14,317	565.60	317.63	0.00	1,594.76
Net Price ($)	14,317	29,122.77	13,216.33	3,118.39	111,983.60
Net Price per kW ($/kW)	14,317	3,230.14	970.62	834.28	6,976.19
SMART Rate ($/kWh)	14,317	0.07	0.04	0.00	0.20
Tilt	14,317	29.40	9.48	0	84
Azimuth	14,317	179.86	56.90	0	359
N Inverter	14,317	11.57	13.52	1	78
Annual Generation	14,317	11,289.62	4,428.34	1,547.44	34,383.49
Capacity Factor	14,317	0.14	0.01	0.05	0.17

Subsidies are present values at an 8% discount rate with 0.5% annual panel degradation. Solar Carve-Out subsidies use the ACP–auction midpoint price path.

The distribution of subsidies also differs. In absolute terms the dispersion is similar. Every Solar Carve-Out system received at least $757 per kW, while some SMART systems

received nothing because the value of energy had caught up with the base rate. The maximum subsidy is $2,632 per kW under Solar Carve-Out and $1,595 per kW under SMART, so the two distributions barely overlap. As a result, a comparison of the two programs is a comparison of very different parts of the subsidy range.

Despite the large difference in subsidy size, both programs supported installations with similar technical characteristics. Average system sizes are comparable: 7.96 kW DC under Solar Carve-Out and 9.22 kW DC under SMART. The mean capacity factor is 0.14 in both samples. The Solar Carve-Out sample has 25,491 systems and the SMART sample 14,317.

4 Empirical Strategy

4.1 The Naive Model

I estimate the pass-through of solar subsidies to net installation prices with the regression

$$y_i = \alpha + \beta \text{Subsidy}_i + X_i\gamma + \delta_{u(i)} + \delta_{t(i)} + \varepsilon_i$$

where y_i is the net installation price per kW after subsidy (gross price minus subsidy, per kW of DC capacity), Subsidy_i is the net present value per kW of the subsidy accumulated over ten years, X_i is a set of controls, $\delta_{u(i)}$ is the fixed effect for the utility $u(i)$ that serves project i, and $\delta_{t(i)}$ is the fixed effect for the year $t(i)$ in which it began operating. Because y is net of the subsidy, the coefficient β equals the negative of the pass-through rate ρ: a coefficient of -1 means that a dollar of subsidy lowers the net price by a dollar (the gross price does not change), and a coefficient of -0.76 means the gross price rises by 24 cents.

The controls X_i include fixed effects for panel manufacturer, inverter manufacturer, installer, and city, all of which affect system prices. Panel and inverter manufacturer fixed effects control for equipment quality, installer fixed effects control for labor costs and installer quality, and city fixed effects control for demographic characteristics and local price

levels. Because SMART rates vary across utility territories, utility fixed effects allow me to identify pass-through from changes in the subsidy within a utility territory, where the compensation structure is the same. Year fixed effects capture the effect of technology and learning by doing on installation costs that is common to a year.

I estimate the model separately for the SMART and Solar Carve-Out samples, using the net present value of the subsidy at a discount rate of 8%. The programs differ in period, in how the subsidy is measured, and in the fixed effects the policy requires, so I let every coefficient differ by program and compare the estimates directly (Appendix B.5 compares this with pooled estimation).[18] All prices and subsidies are in nominal dollars.[19]

I call this model the naive specification because it uses all of the variation in the subsidy, including its movement within years, without separating the part that comes from the timing of the subsidy from the part that comes from differences across roofs. Section 4.3 explains why the timing variation is confounded with installation costs, and Section 6 shows the consequences for the estimates.

4.2 Two Sources of Variation

The subsidy per kW varies for two reasons, which differ in how plausibly they are independent of installation costs.

Timing. Under Solar Carve-Out, the subsidy per kW varies across installation dates through the SREC price path and, in January 2017, through the cut in the SREC factor. Year fixed effects absorb any level shift common to all projects in a year, but not the movement of the price path within a year or the interaction between the annual price path

[18] I do not include utility fixed effects for the Solar Carve-Out sample because SREC prices do not depend on the utility. Adding them leaves the Solar Carve-Out estimate essentially unchanged (Appendix Table 17).

[19] I use nominal dollars for two reasons. First, both are nominal contracts: a SMART rate is fixed in dollars per kWh at approval, SRECs are sold at dollar prices, and a buyer compares the dollar price quoted at purchase with the present value of nominal payments discounted at a nominal rate. Pass-through is a within-period object, the effect of a dollar of subsidy on the price paid on the same date, so there is no economic reason to convert to a common base year. Second, time fixed effects already absorb the economy-wide price level. The choice does affect the naive SMART estimate, because consumer prices rose by about 22% over the SMART sample while rates were stepping down (Section 6.4 and Appendix C.2).

and the timing of payments (a project installed in 2016 receives a different sequence of vintage prices than one installed in 2018). Under SMART, the base rate falls as capacity blocks fill within a utility territory, and the value of energy differs by utility and over time. Utility fixed effects hold the compensation schedule constant, so the variation comes from movement along the schedule over time.

Generation across roofs. Subsidy per kW depends on the electricity a system will produce, so two otherwise identical projects in different locations or with different orientations face different subsidies. This variation exists in both programs and does not depend on the policy calendar, so it cannot move together with installation costs over time.

4.3 The Confounding Timing Variation

Write the gross price of project i as $P_i = c_i + (1-\rho)S_i$, where c_i is the cost-based price, the price the installer would charge without the subsidy, and S_i is the subsidy. The dependent variable is the net price $y_i = P_i - S_i = c_i - \rho S_i$. After the fixed effects and controls are partialled out, the regression coefficient converges to

$$\hat{\rho} \;\to\; \rho - \frac{\mathrm{Cov}(\tilde{c}, \tilde{S})}{\mathrm{Var}(\tilde{S})}, \tag{1}$$

where tildes denote residuals. Any component of cost that moves with the residual subsidy biases the estimate. When the identifying variation comes from timing, $\tilde{c}$ contains the part of the cost trend that the time controls do not absorb. Costs that fall as the subsidy falls ($\mathrm{Cov} > 0$) bias $\hat{\rho}$ down, and costs that rise as the subsidy falls ($\mathrm{Cov} < 0$) bias it up.

The two programs give opposite signs. Under Solar Carve-Out, the SREC price path and installation prices fell together from 2014 to 2018. Under SMART, the rate fell throughout while gross prices rose after 2020 (Figure 4). Equation (1) therefore predicts the sign of the bias for every design that relies on timing: downward under Solar Carve-Out and upward under SMART. Narrowing the window around a rule-based change in the subsidy shrinks

the cost trend within the window but does not remove it, and windows narrow enough to remove it contain few projects and are exposed to bunching before the change. Demand shifts point the same way under SMART: blocks fill faster when demand is strong, so strong demand raises the gross price while moving buyers to a lower block, which also biases the estimate up.

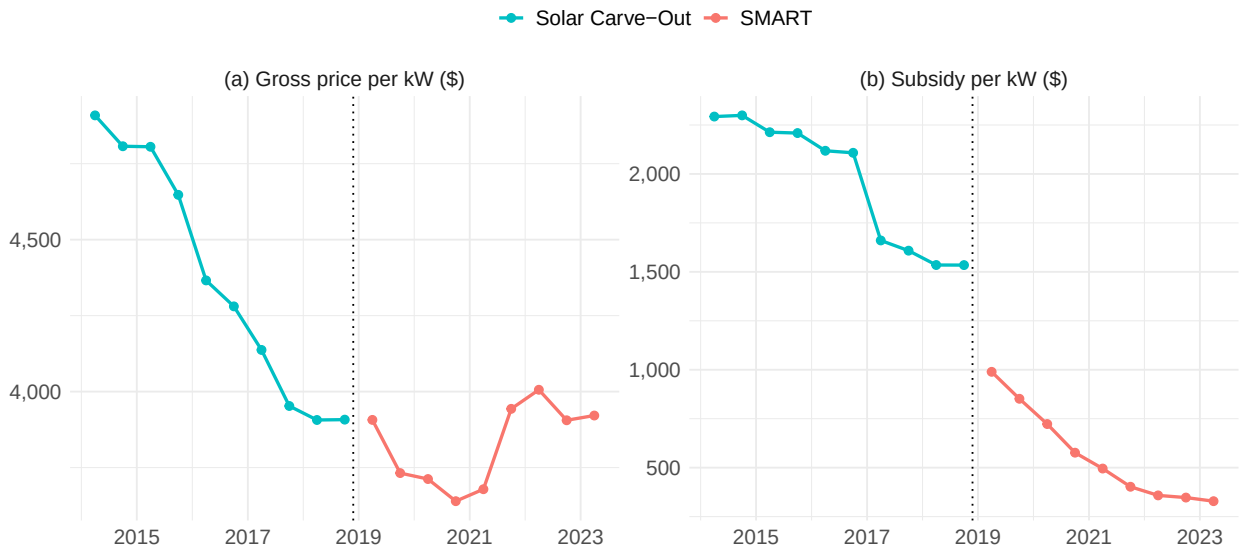


Figure 4: Gross Price and Subsidy per kW over Time. Half-year means of the gross price per kW and the subsidy per kW (present value at an 8% discount rate) by program, without controls. Half-years with fewer than 100 projects are omitted. The dotted line marks the start of SMART.

4.4 The Cross-Roof Design

The preferred specification removes the timing of the subsidy. It replaces year fixed effects with year-month fixed effects (utility-by-year-month for SMART), which absorb the SREC price path and the timing of SMART block changes, adds installer-by-year and city-by-year fixed effects, and controls for the log of system size and five bins of the DC-to-AC ratio. The fixed effects are additive, so pass-through is identified from differences in the subsidy that remain after removing installer-by-year, town-by-year, and month effects, which come mainly from differences in modeled generation. A stricter version replaces installer-by-year

with installer-by-month fixed effects, so that only roofs equipped by the same installer in the same month are compared. More than 90% of projects share an installer and month with another project, and the estimates are nearly identical (Section 5.3). Under SMART, part of the remaining variation also comes from differences in incentive rates among projects that began operating in the same month, because the rate is locked at approval, which is usually months before operation. A further specification instruments the subsidy with the project's modeled capacity factor, so that only the variation that comes from the roof's orientation, tilt, and location identifies pass-through.

Applied to this variation, equation (1) shows what remains a concern. With the timing of the subsidy absorbed, $\tilde{c}$ contains only cost differences across roofs that are correlated with generation, such as a flat roof that needs tilted racking. These differences are the threat to the cross-roof design, and Section 7 examines them.

The two designs estimate different objects. The naive specification uses changes in the subsidy common to all buyers, whereas the cross-roof design uses differences across buyers, and the pass-through of a market-wide change can differ from that of a change specific to one buyer (Muehlegger and Sweeney, 2022). Section 8.1 shows that, with log-concave demand and marginal cost that does not fall with volume, market-wide pass-through cannot exceed the pass-through of a roof-specific subsidy. Two further issues affect both designs. The discount rate enters both the dependent variable and the regressor, so the coefficient depends on it mechanically, and I report a range of rates (Appendix B.1). And because installers know the subsidy schedule, a timing-based design cannot separate the response of prices to a subsidy cut from pricing that anticipated the cut.

Inference. I cluster standard errors at the installer level. Installers set prices, the same installer appears in many projects and years, and an installer's pricing errors are likely to be correlated across its projects. There are 385 installers in the Solar Carve-Out sample and 267 in the SMART sample, which is enough for standard asymptotic inference.[20] To

[20] Clustering at the utility level is not appropriate: with only four utility groups in the SMART sample the cluster count is too small, and in the Solar Carve-Out sample there is no utility-based policy variation.

compare the programs, I estimate both in one stacked regression in which the slope, the controls, and every fixed effect are specific to each program, and I cluster by installer across programs. The point estimates equal those from separate regressions, and the standard error of the difference accounts for the 83 installers that operate in both programs, who installed 56% of Solar Carve-Out projects and 69% of SMART projects.

4.5 Measurement Error in the Subsidy

The subsidy is built from modeled generation, so it is measured with error, and the error enters both sides of the regression, because the dependent variable subtracts the measured subsidy. Write the measured subsidy as $\hat{S}_i = S_i + u_i$. With classical error, $\hat{\rho} \to \lambda\rho + (1 - \lambda)$, where $\lambda = \text{Var}(S)/[\text{Var}(S) + \text{Var}(u)]$, so error pulls the estimate toward one rather than toward zero.[21] Errors in the inputs, such as a misrecorded tilt or azimuth or a geocoding error, are of this kind. Losses that PVWatts does not model and that are unrelated to its inputs, such as shading from a neighbor's tree, are not: if true generation equals modeled generation plus an independent error, the estimate is unbiased. An error common to all projects matters little. If PVWatts overstated every project's generation by a factor k, the estimate would converge to $1 - (1 - \rho)/k$, and even the full statewide gap of $k \approx 1.09$ would turn a true rate of 0.84 into 0.85.

The pull toward one is stronger in the cross-roof design than in the naive design, because roofs within the same cells differ less in true generation, so error makes up a larger share of the remaining variation. Error can move an estimate toward one but not past it, whereas the upward timing bias under SMART can. Under Solar Carve-Out, error works in the same direction as the predicted timing bias. Section 6.1 uses these directions to read the gap between the naive and cross-roof estimates, and Section 7.2 uses the installers' production estimates to gauge the size of the error.

Appendix Table 19 shows how the standard errors change under alternative clustering choices. The installer-clustered errors are larger than those clustered by town in both samples, and the conclusions do not change.

[21] Equivalently, in a regression of the gross price on the measured subsidy, the coefficient $1 - \rho$ is attenuated toward zero.

5 Pass-Through under the Two Programs

5.1 Cross-Roof Estimates

Table 4 reports pass-through estimates from specifications that progressively remove the variation in the subsidy that comes from its timing. Column (1) is the naive specification. Column (2) replaces year fixed effects with year-month fixed effects, interacted with utility for SMART. Column (3) adds installer-by-year and city-by-year fixed effects. Column (4) adds the log of system size in kW DC and indicators for five bins of the DC-to-AC ratio and is the preferred specification, because the price per kW falls with system size and depends on how much inverter capacity each kW of panels needs, and both can differ across roofs that generate different amounts. Column (5) instruments the subsidy with the project's modeled capacity factor in the specification of column (4).[22]

Table 4: Pass-Through from Cross-Roof Variation in Generation

	Dependent variable: net price per kW				
	(1)	(2)	(3)	(4)	(5)
Panel A: Solar Carve-Out					
Subsidy per kW	−0.756***	−0.799***	−0.826***	−0.863***	−0.856***
	(0.044)	(0.046)	(0.044)	(0.044)	(0.050)
Panel B: SMART					
Subsidy per kW	−1.017***	−0.887***	−0.839***	−0.847***	−0.894***
	(0.065)	(0.062)	(0.070)	(0.073)	(0.115)
Difference in pass-through (SMART − Solar Carve-Out)	0.260***	0.087	0.013	−0.016	0.038
	(0.090)	(0.080)	(0.080)	(0.085)	(0.114)
Time fixed effects	Year	Year-month	Year-month	Year-month	Year-month
Installer × year, city × year			✓	✓	✓
System size, DC/AC ratio				✓	✓
Subsidy instrumented with capacity factor					✓

Panels A and B report the coefficient on subsidy per kW. Pass-through is minus this coefficient. The row below the panels reports the difference in pass-through between the programs. All columns include installer, inverter, manufacturer, and city fixed effects (and utility fixed effects for SMART), or their interactions as listed. Discount rate 8%. Observations: 25,491 Solar Carve-Out (23,503 in columns 4 and 5) and 14,317 SMART. Standard errors clustered by installer in parentheses. The difference and its standard error come from one stacked regression with program-specific slopes, controls, and fixed effects, clustered by installer across programs. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

In the preferred specification, column (4), pass-through is 0.86 under Solar Carve-Out

[22]The first-stage Wald statistics are 3,661 (Solar Carve-Out) and 1,298 (SMART). Columns (4) and (5) drop 1,988 Solar Carve-Out projects with missing AC capacity.

and 0.85 under SMART. The difference between the programs is −0.02 with a standard error of 0.085. Both estimates are high but significantly below one. With modeled generation, buyers capture about 85 cents of each dollar of subsidy, whether the subsidy is paid through certificates sold at an uncertain market price or through a fixed rate known at approval. Columns (3) and (5) give the same picture, with differences of 0.01 and 0.04. Instrumenting with the capacity factor, which removes the residual variation in SMART rates, gives 0.89 (0.12) under SMART. Measurement error in modeled generation pulls both levels toward one, so true pass-through may be somewhat lower, and Section 7.2 gauges how much.

The estimates are in the range of prior estimates for solar rebate programs, 78% to close to 100% for host-owned systems (Pless and van Benthem, 2019; Gillingham and Tsvetanov, 2019; Dong, Wiser and Rai, 2018). Those estimates rely on the timing of the subsidy, and Section 6 shows that in Massachusetts such estimates move by more than 0.1 in either direction depending on how costs moved relative to the subsidy.

5.2 What the Comparison Rules Out

A finding of no difference is informative only to the extent that it excludes meaningful differences. In the preferred specification, the 95% confidence interval for the difference between the programs is −0.18 to 0.15. It excludes the naive gap of 0.26, so the cross-roof estimates reject the difference that a comparison of the two programs over time would suggest. Column (3), the version with tilt bins (Table 8), and the version with installer-by-month cells give similar intervals. The rejection depends on precision: the instrumental variables estimates in column (5) and the estimate identified from azimuth alone are less precise, and their intervals reach the naive gap, although their point differences are 0.04 and 0.09.[23]

The programs differ in the form of the payment, in its predictability, and in its size

[23] Accounting for installers that operate in both programs barely changes the standard error of the difference in the preferred specification (0.085, compared with 0.084 if the programs were independent) but raises it in the naive specification, from 0.078 to 0.090. The naive gap remains significant ($p = 0.004$).

(Section 2). Each difference could plausibly change pass-through, and none of them changes it by an amount the data can detect. A buyer receives about the same share of a dollar of subsidy whether it arrives through a certificate market or through a fixed rate, and whether the subsidy covers 45% or 15% of the price. Because the two estimates are nearly identical, a single rate summarizes both programs: one slope with program-specific fixed effects gives 0.86. I nonetheless estimate the programs separately. A pooled slope is a variance-weighted average, with about three-quarters of the weight on Solar Carve-Out, and it imposes the equality that the comparison is meant to test. In the naive specification that equality is rejected, and the pooled rate of 0.84 describes neither program. Appendix B.5 reports the pooled estimates.

5.3 Robustness

Appendix B reports further checks, all in the preferred specification. Comparing only roofs equipped by the same installer in the same month, with installer-by-month in place of installer-by-year fixed effects, gives 0.860 (0.046) under Solar Carve-Out and 0.859 (0.077) under SMART, a difference of −0.002 (0.087), and installer-by-town-by-year fixed effects give 0.853 and 0.848 (Appendix Table 18). The Solar Carve-Out estimate barely moves with the assumed SREC price path, from 0.844 with the auction price to 0.878 with the ACP. Both estimates fall as the discount rate rises from 3% to 15%, from 0.890 to 0.821 under Solar Carve-Out and from 0.879 to 0.796 under SMART, and both stay significantly below one at every rate. The change is a mechanical rescaling of the subsidy. Adding utility fixed effects to the Solar Carve-Out model leaves the estimate essentially unchanged, and clustering by town or utility instead of by installer does not change the conclusions. SMART pass-through does not vary with ZIP-code income or installer concentration (Appendix B.8).

6 Subsidy Timing and Cost Trends

Equation (1) predicts that any estimate identified from the timing of the subsidy is biased down under Solar Carve-Out and up under SMART. This section tests the prediction with every timing-based estimate the data allow: the naive specification, a regression discontinuity at the cut in the SREC factor, a stacked event study around the step-downs in SMART rates, and alternative treatments of the price level. Table 5 collects them, ordered by the length of the period over which each compares projects.

Table 5: Pass-Through by Source of Variation and Length of the Comparison Window

Design	Comparison window	Solar Carve-Out	SMART
Cross-roof, preferred	1 month	0.86 (0.04)	0.85 (0.07)
Cross-roof, other versions	1 month	0.76–0.86	0.85–0.86
Year-month fixed effects	1 month	0.80 (0.05)	0.89 (0.06)
Event study, SMART step-downs	8 months		1.27 (0.15)
Regression discontinuity,	±90 days	1.03 (0.10)	
SREC factor cut	±180 days	0.86 (0.07)	
	±365 days	0.71 (0.08)	
excluding	±90 days	0.76 (0.21)	
Dec. 2016–Jan. 2017	±180 days	0.61 (0.10)	
	±365 days	0.54 (0.10)	
Naive, year fixed effects	1 year	0.76 (0.04)	1.02 (0.07)
Predicted sign of the timing bias		−	+

Pass-through estimates with installer-clustered standard errors in parentheses. The comparison window is the period over which a design compares projects. Cross-roof designs compare projects in the same month within installer-by-year (or installer-by-month) and town-by-year cells; the event study compares projects from four months before to three months after a step-down. Sources: Table 4, columns (4), (2), and (1); Tables 8 and 18 for the other cross-roof versions (tilt bins, azimuth only, installer-by-month cells); Table 7 for the event study; Table 6 for the regression discontinuity, which uses the baseline subsidy measure. The predicted sign follows from equation (1) and the co-movement of costs and the subsidy in Figure 4: costs fell with the subsidy under Solar Carve-Out and rose as it fell under SMART. Discount rate 8%.

Figure 5 plots the same estimates against the length of the period over which each design compares projects. The shaded band is the range of cross-roof estimates across the versions in Table 5. Under SMART, the naive estimate and the event study lie above the band and at or above one. Under Solar Carve-Out, the estimates scatter around and below the band,

and the regression discontinuity falls as its window widens.



Figure 5: Pass-Through and the Length of the Comparison Window. Each point is an estimate from Table 5, plotted against the length of the period over which the design compares projects (log scale), with a 95% confidence interval. Cross-roof and year-month estimates compare projects within a month, the event study within eight months, the regression discontinuity within ±90, ±180, and ±365 days, and the naive model within a year. The shaded band is the range of cross-roof estimates: column (4) of Table 4, tilt bins, azimuth only, and installer-by-month cells. The dashed line marks complete pass-through.

6.1 The Naive Gap

Column (1) of Table 4 gives pass-through of 0.76 (0.04) under Solar Carve-Out and 1.02 (0.07) under SMART, a gap of 0.26. Complete pass-through is rejected for Solar Carve-Out but not for SMART. These estimates are robust to the choice of controls: Appendix Table 24 shows that adding equipment, installer, and city fixed effects moves them little. Taken at face value, the naive estimates would suggest that a fixed-rate subsidy is passed through more fully than a certificate subsidy.

The SMART base rate is not tied to installation costs: it steps down on a fixed schedule as cumulative capacity fills the blocks, and the incentive also falls when the value of energy

rises. Because blocks fill over time, however, the subsidy changes during the same period in which installation costs change, and the SREC price path co-moves with market conditions. As Section 4.3 notes, the Solar Carve-Out subsidy and gross price fell together from 2014 to 2018 (Figure 4). Under SMART, the subsidy fell from $989 per kW in the first half of 2019 to $329 in the first half of 2023, while the gross price fell until 2020 and then rose from $3,639 to $4,006 per kW by the first half of 2022, as consumer prices rose. The same pattern holds within years: after removing year means, the monthly mean gross price and subsidy are positively correlated under Solar Carve-Out and negatively correlated under SMART. These are the signs of $\text{Cov}(\tilde{c}, \tilde{S})$ in equation (1) behind the predictions of Section 4.3, and they match the directions in which the naive estimates differ from the cross-roof estimates.

Year-month fixed effects alone do most of the work: the gap is already down to 0.09 in column (2), before any of the other controls are added. As the timing of the subsidy is absorbed, the Solar Carve-Out estimate rises from 0.76 to 0.86 and the SMART estimate falls from 1.02 to 0.84–0.89. Most of the naive gap therefore reflects how each subsidy moved over time relative to costs, not how the two designs divide the subsidy. The two movements are not equally informative. The fall in the SMART estimate goes against the pull of measurement error, so the naive SMART estimate is biased up by at least 0.17, and by more if error inflates the cross-roof estimate. The rise in the Solar Carve-Out estimate goes in the direction measurement error would also produce, and the cross-roof estimate identified from azimuth alone, 0.76, equals the naive estimate. The data therefore do not show that the naive Solar Carve-Out estimate is biased, even though costs fell with the subsidy. The naive gap between the programs comes mainly from the SMART estimate.

6.2 Regression Discontinuity at the SREC Factor Cut

On January 8, 2017, the SREC factor for new small systems fell from 1.0 to 0.8 (Section 2). A system that generates the same electricity receives 20% fewer certificates after the cutoff, so the subsidy per kW falls discretely without any change in the installer's costs. A regression

discontinuity in time at this cut is the design one would use to avoid trends. I use the commercial operation date as the running variable, with a linear trend on each side of the cutoff and installer, inverter, manufacturer, and city fixed effects, and I instrument the subsidy with the post-cutoff indicator. Figure 6 plots monthly means by month of operation. The subsidy per kW falls from about $2,110 in December 2016 to about $1,610 from February 2017, and every project operating after the cutoff has a factor of 0.8, with a single exception.

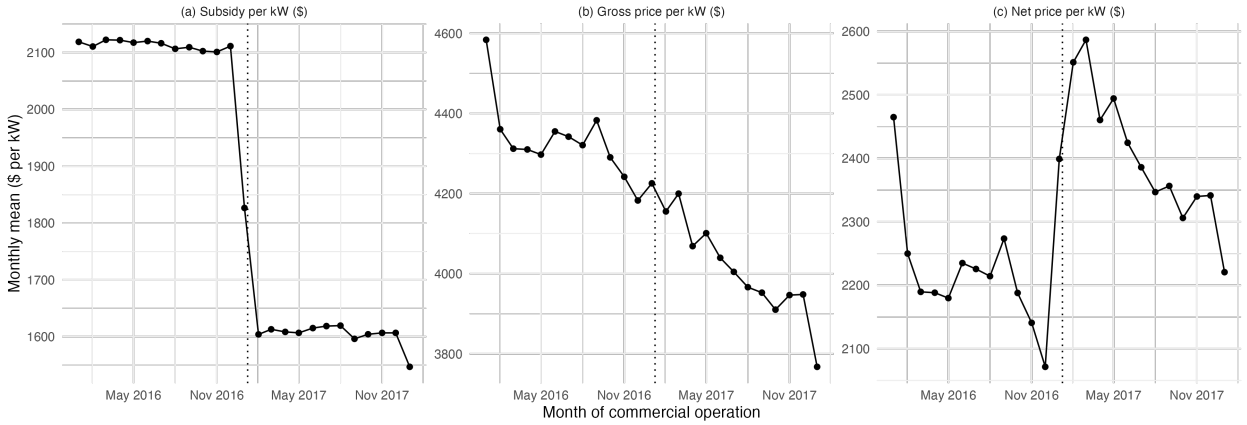


Figure 6: Subsidy, Gross Price, and Net Price by Month of Operation around the SREC Factor Cut

Two features of the raw data complicate the comparison. First, gross prices fall throughout 2016–2017, by roughly $230 per kW per year, which is the cost trend that a design in time has to control. Second, the number of projects operating in December 2016 (879) is well above that in each of the eleven preceding months (358–647), which suggests a rush to complete before the deadline, and December projects have unusually low gross prices (about $4,180 per kW, compared with $4,240–$4,380 in the preceding months). I therefore report estimates with and without the two months around the cutoff.

Table 6 reports the estimates. The subsidy falls by $465–$500 per kW in the baseline measure and $420–$435 in the factor-only measure, which holds the SREC price path fixed and isolates the change caused by the factor. The estimate depends on the window. With all projects, pass-through falls from 1.03 at a window of ±90 days to 0.86 at ±180 days and 0.71 at ±365 days. Excluding the two months around the cutoff, it falls from 0.76 to 0.61

and 0.54. The wider the window, the more of the falling cost trend enters the comparison and the closer the estimate comes to the naive 0.76, or below it, which is the direction equation (1) predicts. The narrowest windows are imprecise (standard errors of 0.21–0.25 when the rush is excluded) and cannot reject the cross-roof estimates of 0.76–0.86, and the all-project estimate at ±90 days is pushed up by the low December prices, which widen the apparent jump in net prices. No single estimate is decisive, and the discontinuity does not by itself show the sign of the bias under Solar Carve-Out. The evidence is that a design built to avoid trends still gives an answer that moves with the cost trend in its window.

Table 6: Discontinuity at the January 2017 SREC Factor Cut

	All projects			Excluding Dec. 2016–Jan. 2017		
Bandwidth (days)	±90	±180	±365	±90	±180	±365
	(1)	(2)	(3)	(4)	(5)	(6)
Subsidy per kW, baseline measure (first stage)	−466***	−478***	−478***	−495***	−497***	−489***
	(12)	(9)	(8)	(34)	(13)	(10)
Subsidy per kW, factor-only measure (first stage)	−434***	−430***	−424***	−425***	−427***	−422***
	(11)	(9)	(8)	(34)	(13)	(11)
Gross price per kW	16	−65**	−139***	−117	−192***	−227***
	(46)	(33)	(37)	(104)	(50)	(46)
Net price per kW	482***	413***	339***	378***	305***	263***
	(48)	(34)	(41)	(114)	(51)	(49)
Pass-through $\hat{\rho}$, baseline measure	1.034***	0.864***	0.708***	0.763***	0.613***	0.537***
	(0.098)	(0.068)	(0.079)	(0.213)	(0.100)	(0.096)
Pass-through $\hat{\rho}$, factor-only measure	1.110***	0.960***	0.799***	0.888***	0.713***	0.622***
	(0.105)	(0.076)	(0.089)	(0.246)	(0.117)	(0.110)
Observations	3,206	5,995	11,795	1,722	4,527	10,328

Solar Carve-Out projects. Each column is a set of regressions on projects within the stated window around January 8, 2017, using commercial operation date as the running variable, with a linear trend on each side of the cutoff and installer, inverter, manufacturer, and city fixed effects. Entries for the first four rows are the estimated jump at the cutoff in dollars per kW. Pass-through is estimated by two-stage least squares with the post-cutoff indicator as the instrument for the subsidy. The factor-only measure holds the present-value factor of the SREC price path at its 2016 value, so that the subsidy varies only through the SREC factor and generation. Standard errors clustered by installer in parentheses. Discount rate 8%. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

6.3 Event Study around SMART Step-Downs

The same logic applies to SMART: when a capacity block fills, the base rate for the utility falls by a known amount, and the subsidy per kW falls discretely. I identify step-downs from

the monthly modal SMART rate: a step is a month in which the modal rate falls by more than \$0.004 per kWh. Reading the rate this way requires many projects in each month, so I use the two utilities that account for 90% of SMART projects, Eversource East (7,400 projects, 51 months with at least 15 projects) and National Grid (5,747 projects, 46 such months). Unitil has 88 projects and never reaches 15 in a month. Eversource West has 1,082 projects, and its modal rate is noisy in the months with few installations. Adding it gives twelve steps and a pass-through estimate of 1.26 (0.15), almost identical to the estimate below. I keep nine steps that are at least four months from any other step. Each step lowers the rate by \$0.011–\$0.020 per kWh. I stack the nine events and estimate the change in subsidy, gross price, and net price in the four months before and three months after each one, relative to the month before the step.

The data record the date a project began operating, while the SMART rate is fixed when the application is approved, which is usually months earlier. The subsidy therefore does not jump at the step but declines gradually across the window (Figure 7, panel a): it is about \$51 per kW above its month −1 level four months before the step and \$88 below it three months after. If pass-through is complete, the gross price should not respond, and the net price should rise by the amount the subsidy falls. Panel (b) shows that the gross price does not break at the step. Its event-time coefficients lie between −\$40 and +\$26 per kW, with confidence intervals of roughly ±\$40, and panel (c) shows that the net price rises as the subsidy falls.

Table 7 pools the post-step periods. The subsidy is \$85 per kW lower after a step, the gross price is \$23 higher (a standard error of \$13), and the net price is \$108 higher, which implies pass-through of 1.27 (standard error 0.15) when the post indicator instruments the subsidy. The estimate is above one, as the naive SMART estimate is, and in the direction equation (1) predicts for a period in which gross prices drifted up as the subsidy fell. Adding a linear trend in event time yields an estimate of 1.65 with a standard error of 1.43, so once the gradual decline in the subsidy is absorbed by the trend, the design does not identify

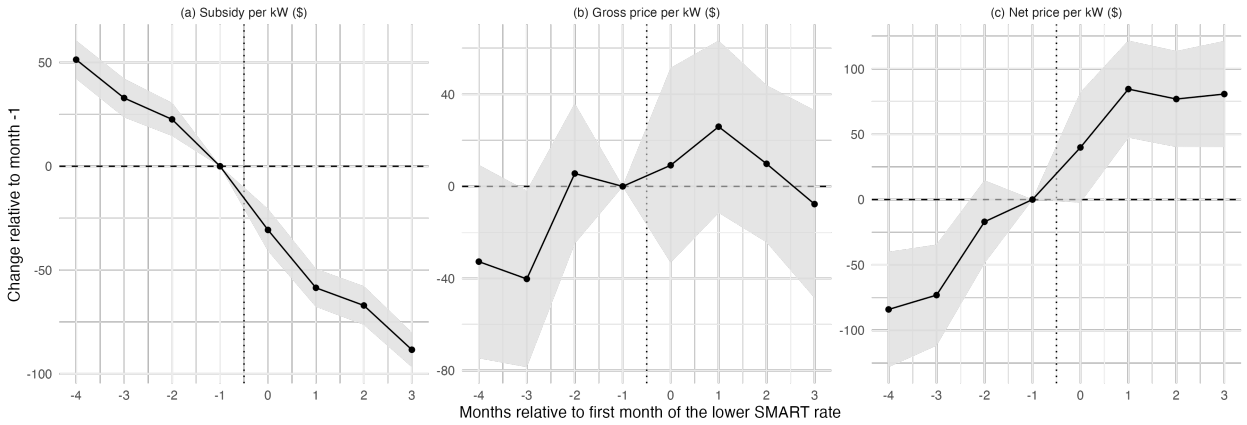


Figure 7: Event Study around SMART Rate Step-Downs

pass-through. Because the rate is locked at approval, the event study compares projects a few months apart rather than across a sharp break. It also rests on nine events in two utilities and cannot rule out that some step-downs coincide with updates to the value of energy.

Table 7: Pooled Event Study around SMART Rate Step-Downs

	No trend	Linear event-time trend
	(1)	(2)
Subsidy per kW, post vs. pre ($)	−85.2***	−18.2***
	(2.4)	(5.8)
Gross price per kW, post vs. pre ($)	23.2*	11.8
	(12.9)	(25.7)
Net price per kW, post vs. pre ($)	108.4***	30.0
	(12.8)	(26.1)
Pass-through, IV	1.272***	1.645
	(0.153)	(1.435)
Observations	10,851	10,851

SMART projects. Stacked sample of nine step-downs in the modal SMART rate (two National Grid, seven Eversource East), with projects from four months before to three months after the first month of the lower rate. Post is the indicator for event time $k \geq 0$, and column (2) adds a linear trend in event time. Regressions include event, installer, inverter, manufacturer, and city fixed effects. Pass-through is estimated by two-stage least squares with the post indicator as the instrument for the subsidy. Standard errors clustered by installer in parentheses. Discount rate 8%. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

6.4 Other Timing Evidence

Three further results point the same way for SMART. First, the naive SMART estimate is sensitive to the treatment of the price level, because consumer prices rose by about 22% over the SMART sample while rates were stepping down. It is 1.02 in nominal dollars, 0.91 with the CPI as a control, 0.93 with year-by-month fixed effects, and 0.85 in CPI-deflated dollars, while the Solar Carve-Out estimate stays between 0.73 and 0.80 (Appendix C.2). Every alternative moves the SMART estimate toward the cross-roof estimate, because each removes part of the co-movement between the subsidy and costs. Second, by quartile of the subsidy, the naive SMART estimates are 1.55 and 1.32 in the two lowest quartiles, where the subsidy is smallest and the projects are latest (Appendix Table 26). Third, in the naive specification SMART pass-through falls with ZIP-code income, from 1.09 to 0.91 ($p = 0.011$), and this gradient disappears once the timing of the subsidy is absorbed (Appendix B.8).

6.5 Market-Wide and Roof-Specific Pass-Through

The timing-based designs and the cross-roof design estimate different objects, so a difference between them need not reflect bias. Under SMART, the timing-based estimates exceed the cross-roof estimate, and with log-concave demand and marginal cost that does not fall with volume, market-wide pass-through cannot exceed the pass-through of a roof-specific subsidy (Section 8.1). Estimates of 1.02 and 1.27 therefore cannot reflect the difference in estimands. Under Solar Carve-Out, the naive estimate lies within the range of cross-roof estimates (0.76–0.86), and the regression discontinuity estimates scatter on both sides of it, so there is no gap for the difference in estimands to explain. A lower market-wide rate would be consistent with a rising supply curve, but under Solar Carve-Out cross-roof pass-through does not fall when an installer or the state is busy (Section 8.4). Appendix C.3 considers three further explanations for the naive gap, the size of the subsidy, its predictability, and competition, and finds that none of them can be separated from the timing of the two programs.

6.6 Implications for Existing Estimates

The solar pass-through estimates in the literature use the kind of variation that this section shows to be confounded in Massachusetts. Pless and van Benthem (2019) identify pass-through from "large and sudden jumps in CSI rebate levels" relative to smooth time trends, and their host-owned estimate ranges from 0.70 to 1.01 across the trend specifications in their Table 3. Adding utility-specific trends alone moves it from 0.78 to 1.01, and complete pass-through can then no longer be rejected. Dong, Wiser and Rai (2018) use rebate step-downs within and across California's utilities with ZIP and month fixed effects, Gillingham and Tsvetanov (2019) use Connecticut's statewide step-downs with year fixed effects, and Podolefsky (2013) compares systems six months before and after one change in the federal tax credit with a cubic trend. In each, rebate schedules that fall as deployment grows make the subsidy move with installation costs, and equation (1) applies. I do not re-estimate these studies, and their estimates may well be close to the truth. The point is that their identifying variation, unlike variation across roofs at the same time, does not separate how the subsidy is shared from how costs moved.

7 Threats to the Cross-Roof Design

The cross-roof design rests on one assumption: within an installer, town, and period, generation affects the gross price only through the subsidy. This section examines the three ways it could fail.

7.1 Roof Traits and Installation Cost

Roofs that generate more could cost more, or less, to install, in ways that system size and the DC-to-AC ratio do not capture, such as pitch, access, or racking. If installation cost changes by c dollars per kW for each percentage point of capacity factor, and the subsidy rises by s, equation (1) gives $\hat{\rho} = \rho - c/s$. Within the cells of the preferred specification, the

subsidy rises by $131 per kW per percentage point under Solar Carve-Out and by $40 under SMART, so a cost difference common to both programs would move the SMART estimate about 3.3 times as much as the Solar Carve-Out estimate and would open a gap between them. The equal estimates already argue against a large common c.

Within cells, the capacity factor varies mainly with the direction the roof faces. Azimuth bins explain 83% (Solar Carve-Out) and 89% (SMART) of the residual variance of the capacity factor, and tilt bins 9% and 3%. Azimuth determines where the panels point, not how much work the installation takes. Tilt can change cost: flat roofs need tilted racking and steep roofs need more safety work. Appendix Table 21 reports the gross price by tilt bin, holding the subsidy fixed. The only sizable difference is for nearly flat roofs under Solar Carve-Out, which cost $372 (122) more per kW than roofs pitched at 25–30 degrees and generate less. Because those roofs cost more and generate less, they bias the estimate up.

Table 8 removes these differences. Holding tilt fixed with five-degree bins lowers the Solar Carve-Out estimate from 0.86 to 0.82, as the flat-roof premium predicts, and leaves SMART at 0.85. The difference between the programs becomes 0.03 (0.08), with a 95% interval of −0.13 to 0.18. I keep column (4) as the preferred specification because it uses all of the within-cell variation in modeled generation, including the part that comes from tilt; the tilt-bin version shows that the flat-roof premium accounts for about 0.04 of the Solar Carve-Out estimate. Identifying pass-through only from azimuth, by instrumenting the subsidy with azimuth bins, gives 0.76 (0.04) under Solar Carve-Out and 0.86 (0.12) under SMART, a difference of 0.09 (0.12) that is not significant. Under SMART, every version gives 0.85–0.86. Under Solar Carve-Out, the estimates range from 0.76 to 0.86 depending on which roof traits identify pass-through, so the conclusion that the two programs pass through the same share holds in every version, but the Solar Carve-Out level is less certain than its standard error suggests. The lowest value, from azimuth alone, is what one would expect if part of the remaining variation in modeled generation were error in the other inputs, such as tilt and location. Azimuth is recorded directly on the application, and instrumenting with it

removes error in the other inputs that is unrelated to azimuth. Section 7.2 shows that error of this size is plausible.

Table 8: Cross-Roof Pass-Through and Roof Geometry

	Dependent variable: net price per kW		
	Cross-roof (preferred) (1)	Tilt bins added (2)	Azimuth only (IV) (3)
Panel A: Solar Carve-Out			
Subsidy per kW	−0.863***	−0.820***	−0.762***
	(0.044)	(0.035)	(0.039)
First-stage F statistic			2942.6
Observations	23,503	23,503	23,503
Panel B: SMART			
Subsidy per kW	−0.847***	−0.845***	−0.855***
	(0.072)	(0.072)	(0.121)
First-stage F statistic			253.6
Observations	14,317	14,317	14,317
Difference in pass-through (SMART − SCO)	−0.016	0.026	0.093
	(0.085)	(0.080)	(0.116)

Column (1) is column (4) of Table 4. Column (2) adds five-degree tilt bins, specific to each program. Column (3) instruments the subsidy with fifteen-degree azimuth bins in the specification of column (2), so that only azimuth identifies pass-through. All columns come from one stacked regression with program-specific slopes, controls, and fixed effects, clustered by installer across programs. The first-stage F statistics are computed under homoskedasticity. Discount rate 8%. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

7.2 Measurement Error in Modeled Generation

As Section 4.5 shows, error in modeled generation pulls the estimates toward one, and more so in the cross-roof design. Instrumenting the subsidy with the modeled capacity factor (column 5 of Table 4) does not remove this, because the instrument comes from the same PVWatts run.

The installer's production estimate on each SMART application gives a sense of how large this pull is (Table 9). Within the cells of the preferred specification, the two measures of the capacity factor have a correlation of only 0.57, so a substantial part of the remaining variation in modeled generation may be error. Computing the subsidy from the installer's

estimate instead of PVWatts gives SMART pass-through of 0.77 (0.07), compared with 0.85 with PVWatts on the same projects, and instrumenting the PVWatts subsidy with the installer-estimate subsidy gives 0.74 (0.08). Complete pass-through is rejected in every version.

I do not use these numbers as the main estimates, for three reasons. First, the installer's estimate is prepared for the sale, with tools and loss assumptions that differ across installers, and its error can be correlated with the price, which would bias both versions. An installer that overstates output to make a system look attractive may also charge more for it, which would push both estimates down. Second, the instrumental variables version requires the errors of the two measures to be independent, which this correlation would violate. Third, Solar Carve-Out records include no production estimate, so the alternative measure cannot be built consistently for both programs. PVWatts applies one model and one set of loss assumptions to every roof in both programs, which is what the comparison between them requires.

The sensitivity check therefore bears on the level of pass-through, not on the comparison. Error in modeled generation pulls the estimates toward one, so true pass-through may be lower than 0.85, which strengthens the conclusion that it is incomplete. Because both subsidies are built from the same model, the error has the same source in both programs. Under Solar Carve-Out, the estimate identified from azimuth alone, which is less exposed to error in the other inputs, is 0.76 (Table 8), close to the SMART values with the installer's estimate. Section 8.3 shows how the bound on market power changes with the amount of error.

7.3 Rates Locked at Approval

Under SMART, the rate is locked when the application is approved, while the data record the month the system began operating. Projects that began operating in the same month in the same utility can therefore carry different rates, so part of the SMART variation in

Table 9: Modeled Generation and Installers' Production Estimates under SMART

Panel A: Agreement of the two measures (SMART)	Capacity factor	Subsidy per kW
Correlation, all projects	0.472	0.965
Correlation within cells of column (4)	0.565	0.831
Reliability of PVWatts within cells, λ	0.570	0.795
Panel B: SMART pass-through by measure of generation		
	$\hat{\rho}$	SE
PVWatts (column 4, validation sample)	0.846	(0.076)
Installer's production estimate	0.773	(0.069)
PVWatts, instrumented with the installer's estimate	0.739	(0.079)

Panel A compares PVWatts output with the estimated annual production reported on each SMART application, for the 14,196 projects (99%) whose estimate lies within 50% of the PVWatts value. Within-cell values use residuals after the fixed effects and controls of column (4) of Table 4. $\lambda = \text{Cov}(r_{PV}, r_I)/\text{Var}(r_{PV})$, where r_{PV} and r_I are the residuals of the PVWatts and installer measures, is the share of the within-cell variation in the PVWatts measure that is true variation if the errors of the two measures are independent. The subsidy per kW varies with the SMART rate as well as with generation. In Panel B, the installer-estimate subsidy uses the same rate and present-value factor with the installer's estimate of generation. The instrumental variables estimate instruments the PVWatts subsidy with it (first-stage Wald statistic 1,371). Both alternative estimates are sensitivity checks: the installer's estimate is prepared for the sale and its error may be correlated with the price. Standard errors clustered by installer in parentheses. Discount rate 8%.

column (4) comes from rates rather than generation. These rate differences reflect when a project applied, which could be correlated with its cost. Column (5) of Table 4 uses only the variation in the subsidy that comes from the modeled capacity factor and gives 0.89 (0.12), close to column (4). The SMART estimate therefore does not depend on the residual rate variation.

8 Implications for Market Power

High pass-through is often read as evidence of competition, but the two are not the same (Weyl and Fabinger, 2013; Pless and van Benthem, 2019). This section asks what the cross-roof estimates reveal about installers' market power. The answer is that the market power the subsidy reveals is a limited amount of pricing by roof, which bounds markups under SMART at below about 10% of price in most specifications, and that the other sources of market power the data can measure leave little trace.

8.1 Framework

Weyl and Fabinger (2013) provide a framework for interpreting the size of pass-through. Consider a symmetric market in which installers' conduct is summarized by a parameter θ, which equals 0 under perfect competition and 1 under monopoly and does not change with the subsidy. The pass-through rate of a per-unit subsidy that changes for all buyers at once is

$$\frac{1}{\rho_M} - 1 = r + \frac{\theta}{\varepsilon_{ms}}, \tag{2}$$

where $r \geq 0$ is the slope of marginal cost relative to the slope of inverse demand and ε_{ms} is the elasticity of the marginal consumer surplus, which measures the curvature of demand. The term $1/\varepsilon_{ms}$ is zero when demand is log-linear, positive when demand is log-concave (for example, linear), and negative when it is log-convex (for example, constant elasticity).[24] With log-concave demand, market power pushes pass-through below one, so a market-wide pass-through rate above one requires marginal cost that falls with volume.

The cross-roof estimates use a different kind of variation: differences in the subsidy across buyers of the same installer in the same town and month. A higher subsidy for one buyer does not change the installer's volume appreciably, so the supply term drops out, and the pass-through of a roof-specific subsidy is

$$\frac{1}{\rho_B} - 1 = \frac{\theta_B}{\varepsilon^B_{ms}},$$

where θ_B measures the installer's pricing power over individual buyers and ε^B_{ms} is the curvature of demand among buyers with similar roofs (Appendix E derives this expression from a pricing condition for each roof type). With log-concave demand and $r \geq 0$, $\rho_M \leq \rho_B$. If installers charged one price per kW regardless of the roof, or if the market were perfectly competitive, then $\rho_B = 1$, however large the installer's markup. Pass-through below one

[24] In elasticities, $1/\rho_M = 1 + (\varepsilon_D - \theta)/\varepsilon_S + \theta/\varepsilon_{ms}$, where ε_D is the absolute value of the demand elasticity and ε_S the elasticity of supply. Pless and van Benthem (2019) use the same expression, with an additional term for conduct that changes with the subsidy.

therefore requires that the gross price rise with the roof's subsidy: the shortfall $1 - \rho_B$ is the share of a roof's subsidy that the installer captures by pricing the roof. The distinction parallels Muehlegger and Sweeney (2022), who show that fine fixed effects identify the pass-through of a shock specific to one firm, which can differ greatly from that of a shock common to the industry. A roof-specific subsidy differs from a firm-specific cost shock in one respect: every installer that quotes the roof faces the same subsidy, so competition for the job already reflects it.

8.2 Price Discrimination

Pricing the roof is a form of price discrimination, and the conditions for it hold unusually well in this market. Installers quote each project individually, and they compute the roof's expected output themselves: every SMART application reports the installer's estimate of the system's annual production (Section 3.4), so the installer knows the subsidy the buyer will receive. An installation cannot be resold. A production-based subsidy also makes such pricing easier than an upfront rebate does, because it varies with a characteristic the installer observes. An installer with market power would therefore have both the means and the incentive to charge more for roofs that earn more.

Installers do set individual prices, but they price the roof only a little. Table 10 compares the dispersion of prices with the price gap that roof productivity produces. After the controls of column (4) of Table 4, the gross price per kW varies with a standard deviation of \$447 under Solar Carve-Out and \$488 under SMART among buyers of the same installer in the same month, so prices are not set from a uniform schedule. A roof that is one standard deviation more productive earns \$137 more subsidy per kW under Solar Carve-Out and \$59 more under SMART, but it is charged only \$20 and \$6 more, about 4% and 1% of the residual dispersion of prices. The ratio of the two gaps is $1 - \rho$ in column (5) of Table 4. Pass-through is also nearly the same on more and less productive roofs (Appendix Table 22), and Appendix F discusses the welfare effects of pricing by roof.

Table 10: Price Dispersion and Pricing on Roof Productivity

	Solar Carve-Out	SMART
Mean gross price per kW ($)	4,270	3,796
Residual SD of gross price per kW ($)	485	521
within installer and year-month ($)	447	488
SD of modeled capacity factor (pp)	1.04	1.45
Subsidy gap for a 1-SD more productive roof ($/kW)	137	59
Gross-price gap for a 1-SD more productive roof ($/kW)	20	6
	(7)	(7)
Gross-price gap as % of residual SD	4.1%	1.2%

Residual standard deviations of the gross price per kW after installer-by-year, city-by-year, year-month (utility-by-year-month for SMART), inverter, and manufacturer fixed effects, log system size, and DC-to-AC ratio bins. The within installer and year-month row uses projects that share an installer and month with at least one other project. The gaps are the coefficients on the modeled capacity factor (percentage points) in regressions of the subsidy per kW and of the gross price per kW with the same controls, multiplied by the standard deviation of the capacity factor. Discount rate 8%. Standard errors clustered by installer in parentheses.

8.3 Bound Market Power under SMART

Turning the shortfall into a bound on conduct requires the curvature of demand. Following Pless and van Benthem (2019), I estimate demand at the ZIP-code and year level, with installations per 1,000 residents quadratic in the mean net price per watt, ZIP characteristics, and county and year fixed effects, and I instrument the price with the mean subsidy per watt and the county wage in building-equipment contracting (Appendix D.1). Under SMART, demand slopes down, with an elasticity at the mean price of −1.57 in the quadratic form and −2.16 in the log-quadratic form, and it is log-concave: the curvature $1/\varepsilon_{ms}$ is 5.38 with the quadratic in levels and 2.14 with the log-quadratic form, which rejects log-linear demand. Both curvature estimates are only marginally significant (t-statistics of about 1.9 and 2.1), and the bound below depends on their sign. Under Solar Carve-Out, demand is not identified, because the subsidy is driven by the statewide certificate price and varies little across ZIP codes within a county and year. I therefore draw structural conclusions only for SMART.

The subsidy instrument raises a concern that Section 6 makes concrete. SMART rates

step down as capacity blocks fill, and blocks fill faster where demand is strong, so the mean subsidy in a ZIP code and year can be correlated with demand shocks. Two checks bear on this. County-by-year fixed effects absorb demand shocks common to a county and year, including most of the utility-wide shocks that move blocks, and with them the log-quadratic curvature is almost unchanged (2.13, standard error 1.12; Appendix D.2). An instrument built only from roof geometry, the mean modeled capacity factor of a ZIP code's projects in a year, shifts the subsidy without the rate, but it is too weak to be informative (first-stage Wald statistics below 10), so it cannot replace the subsidy instrument. The bound is therefore conditional on the demand estimates.

Because r is absent, $\theta_B = (1/\rho_B - 1)\,\varepsilon_{ms}$ holds with equality if all of the shortfall below one reflects pricing, and it is an upper bound if part of the shortfall reflects cost differences across roofs. The Lerner index equals θ/ε_D. Table 11 reports the bound on markups for three values of ρ_B and six demand estimates. With modeled generation ($\rho_B = 0.847$, Table 4, column 4), θ is at most 0.034 with the quadratic demand and 0.085 with the log-quadratic demand, so installer markups are at most 2.1% and 3.9% of price, and 4.1–7.0% with the flexible forms of Appendix D.3. If pass-through is instead measured with the installer's production estimate, at 0.77 or 0.74 (Section 7.2), the shortfall below one is half again as large or more, and the bound rises to 3.5–11.3% and 4.2–13.7% of price. Markups stay below 10% of price in 15 of the 18 combinations. The calculation uses the curvature of aggregate ZIP-level demand in place of the curvature among buyers with similar roofs. Sampling uncertainty widens the upper end: drawing ρ and $1/\varepsilon_{ms}$ from their estimated distributions at $\rho_B = 0.847$, the 95th percentile of the bound on θ is 0.15 with the quadratic demand and 0.35 with the log-quadratic demand, which correspond to markups of about 10 and 16 percent of price.

Table 12 shows the same bound as a function of the share λ of the remaining variation in the measured subsidy that is true variation. If λ is known, $\rho_B = 1 - (1 - \hat{\rho}_B)/\lambda$. Markups exceed 5% of price only if λ is below 0.82 with the log-quadratic demand or 0.52 with the quadratic demand, and 10% only if it is below 0.48 or 0.33. Taken at face value, the installers'

Table 11: Upper Bound on Installer Markups under SMART

	Roof-specific pass-through ρ_B		
	0.847	0.773	0.739
Demand estimate	Modeled generation	Installers' estimates	
Quadratic, 2SLS	2.1	3.5	4.2
Log-quadratic, 2SLS	3.9	6.3	7.7
Quadratic, control function	5.5	8.8	10.7
Natural spline, 3 df	4.6	7.4	8.9
Natural spline, 4 df	4.1	6.6	8.0
Natural spline, 5 df	7.0	11.3	13.7

Entries are the upper bound on the markup in percent of price, $(1/\rho_B - 1)/(1/\varepsilon_{ms})$ divided by the absolute elasticity of demand, at the point estimates of the curvature and elasticity at the mean SMART price (Tables 27 and 29). ρ_B is column (4) of Table 4, with modeled generation, and the two sensitivity values of Table 9, with the installers' production estimates (OLS and instrumental variables).

Table 12: Bound on Market Power under Measurement Error in Generation

		Quadratic demand		Log-quadratic demand	
λ	ρ_B	θ bound	Markup (% of price)	θ bound	Markup (% of price)
1.0	0.847	0.034	2.1	0.085	3.9
0.9	0.830	0.038	2.4	0.096	4.4
0.8	0.808	0.044	2.8	0.111	5.1
0.7	0.781	0.052	3.3	0.131	6.1
0.6	0.745	0.064	4.1	0.161	7.4
0.5	0.693	0.082	5.2	0.207	9.6
λ at which the markup reaches 5%: 0.52 (quadratic), 0.82 (log-quadratic)					
λ at which the markup reaches 10%: 0.33 (quadratic), 0.48 (log-quadratic)					

λ is the share of the variation in the measured subsidy, within the cells of the preferred specification, that reflects true differences in generation. $\rho_B = 1-(1-\hat{\rho}_B)/\lambda$ with $\hat{\rho}_B = 0.847$ (Table 4, column 4). The bound is $\theta \leq (1/\rho_B - 1)\,\varepsilon_{ms}$ and the markup is θ/ε_D, at the point estimates of the SMART demand curvature and elasticity in Table 27. Taken at face value, the installers' production estimates in Section 7.2 would put λ at about 0.6–0.8.

estimates in Section 7.2 would put λ at about 0.6–0.8, where markups are 3–7% of price.

The same framework gives the division of surplus. The ratio of the gain in consumer surplus to the gain in producer surplus is $I = \rho/[1-(1-\theta)\rho]$, so the buyers' share of the surplus that the subsidy creates is $I/(1+I) = \rho/(1+\theta\rho)$, which equals ρ under perfect competition. The formula requires market-wide pass-through, and because $\rho_M \leq \rho_B$, evaluating it at ρ_B gives an upper bound on the buyers' share. At $\rho_B = 0.847$, buyers receive at most 85% of the surplus if the market is perfectly competitive and at most 82% and 79% at the bounds on θ of 0.034 and 0.085. Market power at the bound therefore shifts at most a few percentage points of the surplus from buyers to installers, beyond what the shortfall in pass-through already shows. The bound assumes that conduct does not change with the subsidy and that marginal cost does not fall with volume. Economies of scale would loosen it.

The comparison with Pless and van Benthem (2019) is direct, because the demand specification is the same. In California, pass-through of 153% to lessees together with convex demand revealed market power in the third-party-owned segment, while host-owned buyers received 78% of the subsidy. In Massachusetts, host-owned buyers receive about 85% of the subsidy under both programs, and log-concave demand turns pass-through below one into an upper bound on market power rather than evidence of it. The two results concern different segments, periods, and states, and they are not in conflict.

8.4 Other Sources of Market Power

Pricing by roof is one way installers could earn a markup. Others would leave different traces, and Table 13 tests their predictions, with the aim of rejecting them. A uniform markup sustained by few competitors would raise gross prices where installers face fewer competitors and, with log-concave demand, lower pass-through where competition is weaker. Rising marginal cost, which lowers market-wide pass-through without market power, implies lower pass-through when installers are busy. Discrimination on willingness to pay rather

than on the roof would make pass-through vary with buyer income.[25]

In the preferred specification, cross-roof pass-through does not fall when an installer is busy: it rises by 0.036 (0.022) per standard deviation of the installer's workload under Solar Carve-Out and by 0.039 (0.025) under SMART. Because a single roof does not move an installer along its marginal cost curve, these interactions test whether busy installers price roofs differently, not whether marginal cost rises, and they do not. Under SMART, pass-through is lower in months of high statewide volume, by 0.073 (0.034) per standard deviation, so installers price the roof somewhat more when demand across the state is strong. Under Solar Carve-Out, statewide volume has no effect (0.014, standard error 0.024). SMART pass-through rises with town concentration, by 0.099 (0.047) per standard deviation, which is the opposite of what market power with log-concave demand would predict, and under Solar Carve-Out it does not change with concentration. Across towns, gross prices are lower where more installers are active, by \$66 per kW per log point ($p = 0.05$), or 1.7% of the mean gross price of \$3,796 per kW. This is small, and it may also reflect demand, since installers enter towns where demand is growing. In the preferred specification, SMART pass-through is 0.80, 0.88, and 0.86 from the least to the most concentrated tercile of ZIP-level installer concentration ($p = 0.22$) and 0.84, 0.85, and 0.83 across terciles of ZIP-code income ($p = 0.84$; Appendix B.8). These are correlations rather than causal estimates, because workload and the number of competitors respond to demand, and the concentration measures vary within a range of low concentration (Appendix C.3.3).

Pass-through cannot detect one source of market power: a markup common to all buyers in a market, such as rents that free entry dissipates through customer-acquisition costs. Under constant marginal cost and log-linear demand, such a markup is a fixed dollar amount that does not respond to the subsidy, and pass-through is one even though installers gain

[25] I measure an installer's workload as the log of its installations in the month of operation and the month before, relative to the installer's own mean, and statewide workload as the log of monthly installations relative to the mean of that year, and I interact each, standardized, with the subsidy. For town competition, I regress the gross price per kW on the log number of installers active in the town and year and, separately, on the town-year HHI, for towns with at least ten installations. Because city fixed effects are included, these comparisons are within a town over time.

Table 13: Workload, Competition, and Pass-Through

	Net price per kW			Gross price per kW	
	Installer workload (1)	Statewide workload (2)	Town HHI (3)	Town competitors (4)	Town HHI (5)
Panel A: Solar Carve-Out					
Subsidy per kW	−0.863***	−0.857***	−0.872***		
	(0.043)	(0.044)	(0.040)		
Subsidy per kW × installer workload	−0.036*				
	(0.022)				
Subsidy per kW × statewide workload		−0.014			
		(0.024)			
Subsidy per kW × town HHI			0.008		
			(0.030)		
Log number of installers in town-year				42.5	
				(31.3)	
Town HHI					−95.7***
					(14.4)
Observations	23,503	23,503	20,086	21,783	21,783
Panel B: SMART					
Subsidy per kW	−0.847***	−0.881***	−0.883***		
	(0.072)	(0.083)	(0.069)		
Subsidy per kW × installer workload	−0.039				
	(0.025)				
Subsidy per kW × statewide workload		0.073**			
		(0.034)			
Subsidy per kW × town HHI			−0.099**		
			(0.047)		
Log number of installers in town-year				−66.1*	
				(34.0)	
Town HHI					−0.8
					(16.3)
Observations	14,317	14,317	11,346	11,346	11,346

The dependent variable is the net price per kW in columns (1)–(3) and the gross price per kW in columns (4) and (5). Columns (1)–(3) use the preferred specification, column (4) of Table 4, so the interactions are identified from differences across roofs within its cells. The main effects of statewide workload and town HHI are absorbed by its fixed effects. Columns (4) and (5) relate prices to the number of installers and the HHI of a town-year, which vary only at the town-year level that city-by-year fixed effects absorb, so they use the naive fixed effects (installer, inverter, manufacturer, city, and year, and utility for SMART) and compare a town with itself over time. Installer workload is the log of the installer's installations in the month of operation and the month before, minus the installer's mean. Statewide workload is the log of all installations in the month, minus the mean for that year. Town competition is measured in town-years with at least ten installations, and the HHI is on a 0–10,000 scale. Workload and HHI are standardized within each program. The coefficient on subsidy per kW is minus pass-through at the mean, and the coefficient on an interaction is minus the change in pass-through per standard deviation. Gross-price regressions also control for subsidy per kW. Discount rate 8%. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

from the subsidy because they sell more systems at a positive markup. The evidence above shows that the market power the subsidy reveals is limited. It does not show that installers earn no markup, which only cost data or a structural model of installer pricing could settle.

8.5 Solar Carve-Out

The two programs pass through the same share of the subsidy across roofs, and 83 installers, accounting for 56% of Solar Carve-Out projects, operate in both. If conduct and the curvature of demand were similar in the two periods, the SMART bound would apply to Solar Carve-Out as well. The data cannot confirm this, because demand under Solar Carve-Out is not identified. The reduced-form checks do not show pricing that responds to capacity: cross-roof pass-through does not fall with installer workload (0.036, standard error 0.022) or statewide workload (0.014, 0.024). The number of competitors in a town has no relation to gross prices, and prices are lower, not higher, in concentrated town-years (−$96 per kW per standard deviation). A likely reason is the Solarize Mass campaigns, in which a town selected a single installer and offered tiered group prices that fell as more households signed up, which made a town concentrated and cheap at the same time (Clean Energy States Alliance, 2015).[26]

9 Conclusion

I estimate the pass-through of two production-based solar subsidies in Massachusetts from differences in the subsidy across roofs that the same installer equips at the same time.

Buyers receive about 85% of the subsidy under both the certificate-based Solar Carve-Out and the fixed-rate SMART program, although one subsidy is more than three times the size of the other and only one is known at the time of sale. The 26-point gap that a

[26]For example, Lexington and Bedford selected Astrum Solar in a campaign that ran until June 2014 (Town of Lexington, 2014), and Astrum, as Direct Energy Solar, is the second-largest installer in the 2015 data.

comparison of the programs over time suggests comes mostly from how each subsidy moved relative to installation costs, not from the design of the payment. Installers raise prices only slightly for roofs that earn larger subsidies, which, with log-concave demand under SMART, bounds their markups at 2–7% of price.

These findings bear on subsidy design. Massachusetts in 2018 and New Jersey in 2021 replaced market-priced certificates with administratively set payments. The results suggest that such a switch changes little about how much of the subsidy reaches buyers, so the choice between the two can rest on other grounds, such as the cost to ratepayers, the price risk that owners bear, and the incentive to generate. Paying for output also lets installers see each roof's subsidy when they quote, yet they price the roof only a little.

They also bear on how subsidies are evaluated. Subsidies for new technologies, such as electric vehicles and batteries, are often scheduled to fall as deployment grows, while costs change with experience and with input prices. Estimates identified from the timing of such subsidies inherit the cost trend, and the direction of the bias depends on how costs moved around each change in the subsidy. Differences in the subsidy across buyers at the same time avoid this problem, and subsidies tied to a characteristic of the buyer, such as a roof's output, often provide them.

These are positive findings about how the subsidy is shared, not a welfare analysis of the two programs, and they come from host-owned residential systems in one state. They also leave a question open. Pass-through identified from differences across buyers shows how far sellers price each buyer's subsidy, which is why it can bound markups without estimating costs, but it cannot detect a markup that installers charge every buyer alike, whether that markup is sustained by limited competition or spent on acquiring customers. Whether such markups make residential solar expensive is a question for data on installers' costs rather than for pass-through.

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A Data Construction

Merging DOER and Tracking the Sun. Projects are identified differently in the two sources. I match them on operation date, size, and total cost simultaneously. This gives an almost perfect one-to-one match, with only a few projects matched to the same Tracking the Sun record, and I verified manually that these projects have the same technical details.

Dates. For the SMART sample, the DOER data list two dates for each project: the Commercial Operation Date and the Incentive Payment Effective Date. I use the Commercial Operation Date, the earliest date in the data. The SMART rate is fixed earlier, when the application is approved, but the data do not record the approval date (Section 7.3).

Geocoding. The PVWatts location input accepts street addresses, ZIP codes, or coordinates. The public DOER data contain only ZIP codes, which can cover large areas relative to the resolution of the solar resource data, so I obtained detailed addresses for each project through a separate data request to DOER. The calculator returned unreliable results for street addresses entered directly, so I converted the addresses to coordinates with the U.S. Census geocoder and used the OpenStreetMap geocoding service for addresses that the Census geocoder could not match. All addresses were geocoded by one of the two services, and I verified that all coordinates lie in Massachusetts.

B Cross-Roof Estimates: Additional Results

B.1 SREC Prices, Discount Rates, and Tests of Complete Pass-Through

SREC prices. I perform a further robustness check specific to Solar Carve-Out using alternative assumptions about system owners' expectations of SREC value. An owner with rational expectations would perfectly predict future SREC prices and sell SRECs once a year at that year's mean price. However, it is hard for an individual to predict the actual price of SRECs, especially in the initial years after the launch of Solar Carve-Out II. The clearinghouse auction price was intended to support SREC prices, but for a period SRECs consistently traded below it because of oversupply (newly installed capacity quickly surpassed the annual target set by the RPS). The common practice of contracting aggregators or brokers to handle SREC trades makes the actual SREC sale price even more difficult to predict. I therefore consider three ways of predicting SREC prices in addition to actual market prices. Optimistic customers use the ACP—the theoretical upper bound—as the predicted SREC price. Pessimistic customers use the clearinghouse auction price. Neutral customers use the average of the ACP and the clearinghouse auction price, which is the main assumption. A standard 10% transaction fee applies to both optimistic and neutral customers. Pessimistic customers are not subject to any other fees, because the transaction fee is already embedded in the clearinghouse auction price.

Table 14 presents pass-through estimates for Solar Carve-Out projects under different SREC price assumptions, in the preferred specification (column (4) of Table 4). Only the Solar Carve-Out sample is shown, and the results are again very close to each other. The highest estimated pass-through rate comes from using the alternative compliance payment as the SREC price, and the lowest from using auction prices. The ordering is as expected, because the ACP is the upper bound of SREC prices and yields the largest subsidy, while the auction price is the lowest assumed path and yields the smallest. The estimates range

from 0.844 to 0.878. The estimate using actual market prices, 0.857, is close to the midpoint estimate of 0.863. Although I cannot determine owners' true beliefs about the income stream SRECs will actually generate, the results show that the Solar Carve-Out estimate is robust to the assumed SREC price path.

Table 14: Pass-Through under Alternative SREC Price Paths

Dependent Variable:	Net Price per kW			
Model:	(1) ACP	(2) Auction	(3) Actual price	(4) Midpoint
Variables				
Subsidy per kW	−0.878***	−0.844***	−0.857***	−0.863***
	(0.039)	(0.050)	(0.046)	(0.044)
Installer × year	✓	✓	✓	✓
City × year	✓	✓	✓	✓
Year-month	✓	✓	✓	✓
Inverter, manufacturer	✓	✓	✓	✓
Log size, DC/AC ratio bins	✓	✓	✓	✓
Observations	23,503	23,503	23,503	23,503

Preferred specification: column (4) of Table 4. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1. Subsidies are discounted at 8% with 0.5% annual panel degradation.

Discount rates. Because the subsidy is a discounted sum of future cash flows, the discount rate affects both the dependent variable (y_i) and the regressor (Subsidy_i). De Groote and Verboven (2019) find that when subsidies are delivered through electricity production, consumers apply high discount rates to future benefits. They estimate an average rate of 15% in Belgium, well above my main assumption and above the contemporaneous risk-free euro interest rate, which remained below 1% during their study period. I therefore also use rates of 3%, 5%, and 15%.

Table 15 presents pass-through estimates under varying discount rate assumptions, in the preferred specification. In Panel A, the estimated pass-through rate declines from 0.890 at a 3% discount rate to 0.821 at 15%, and in Panel B from 0.879 to 0.796. The discount rate mechanically changes the size of the regressor and the dependent variable. For SMART the

change is purely mechanical. The subsidy is $r_i g_i A(d)$ (Section 3.3), so changing d multiplies every subsidy by the same factor. The coefficient on the subsidy in the gross price, $1 - \rho$, is divided by that factor, and $(\hat{\rho}_d - 1)A(d)$ is the same at every discount rate. With $A(8\%)/A(3\%) \approx 0.79$, the preferred estimate implies $\hat{\rho}_{3\%} - 1 \approx -0.153 \times 0.79 \approx -0.121$, which is the estimate in the table, and the test of $\rho = 1$ gives the same p-value (about 0.035) at every rate by construction. The SMART results therefore carry no information about how buyers discount the subsidy. Under Solar Carve-Out the SREC price path differs across cohorts, but the year-month fixed effects absorb these differences, so within a month the discount rate scales every subsidy by nearly the same factor and the result is close to mechanical as well. At every discount rate, both estimates are significantly below one (Table 16).

Table 16 reports the corresponding tests of complete pass-through. For SMART, the standard error times $A(d)$ is also the same at every rate, and so is the p-value. The Solar Carve-Out p-values are nearly constant for the same reason.

Table 15: Pass-Through at Alternative Discount Rates

	Panel A: Solar Carve-Out				Panel B: SMART			
	Dependent Variable: Net Price per kW				Dependent Variable: Net Price per kW			
	(1) d=3%	(2) d=5%	(3) d=8%	(4) d=15%	(5) d=3%	(6) d=5%	(7) d=8%	(8) d=15%
Variables								
Subsidy per kW	−0.890***	−0.880***	−0.863***	−0.821***	−0.879***	−0.867***	−0.847***	−0.796***
	(0.035)	(0.039)	(0.044)	(0.057)	(0.057)	(0.063)	(0.073)	(0.097)
Installer × year, city × year	✓	✓	✓	✓	✓	✓	✓	✓
Inverter, manufacturer	✓	✓	✓	✓	✓	✓	✓	✓
Log size, DC/AC ratio bins	✓	✓	✓	✓	✓	✓	✓	✓
Year-month FE	✓	✓	✓	✓				
Utility × year-month FE					✓	✓	✓	✓
Observations	23,503	23,503	23,503	23,503	14,317	14,317	14,317	14,317

Preferred specification: column (4) of Table 4. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1. 0.5% annual panel degradation. SREC subsidies use the ACP–auction midpoint price path.

B.2 Solar Carve-Out with Utility Fixed Effects

Table 17 adds utility fixed effects to the preferred Solar Carve-Out specification, column (4) of Table 4. SREC prices do not depend on the utility, so the estimate is essentially

Table 16: Test of $\rho = 1$ by Program and Discount Rate

	d=3%	d=5%	d=8%	d=15%
Panel A: Solar Carve-Out				
Subsidy per kW	−0.890***	−0.880***	−0.863***	−0.821***
	(0.035)	(0.039)	(0.044)	(0.057)
p-value, $\rho = 1$	0.002	0.002	0.002	0.002
Panel B: SMART				
Subsidy per kW	−0.879***	−0.867***	−0.847***	−0.796***
	(0.057)	(0.063)	(0.073)	(0.097)
p-value, $\rho = 1$	0.035	0.035	0.035	0.035

Coefficient on subsidy per kW from the preferred specifications, column (4) of Table 4. The dependent variable is the net price per kW, and pass-through ρ is minus the coefficient. The p-values test complete pass-through ($\rho = 1$, a coefficient of −1), using installer-clustered standard errors. Stars test the coefficient against zero. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

unchanged.

Table 17: Solar Carve-Out Pass-Through with and without Utility Fixed Effects

	Preferred	With Utility FE
Subsidy per kW	−0.863***	−0.862***
	(0.044)	(0.044)
Installer × year, city × year FE	✓	✓
Year-month, inverter, manufacturer FE	✓	✓
Log system size, DC/AC ratio bins	✓	✓
Utility FE		✓
Observations	23,503	23,503

Preferred specification: column (4) of Table 4. Standard errors clustered by installer in parentheses. Discount rate 8%. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

B.3 Comparisons within Finer Cells

The preferred specification includes installer-by-year, city-by-year, and month fixed effects additively, so it does not require two compared roofs to share an installer, a town, and a month at once. Table 18 replaces installer-by-year with installer-by-month fixed effects, which compare only roofs that the same installer equipped in the same month, and alternatively with installer-by-town-by-year fixed effects. In the installer-by-month version,

94% of Solar Carve-Out and 92% of SMART projects share an installer and month with another project. The estimates barely move, and the difference between the programs is −0.002 (0.087).

Table 18: Cross-Roof Pass-Through within Finer Cells

	Solar Carve-Out	SMART
Installer × year (column 4)	0.863	0.847
	(0.044)	(0.073)
Installer × year-month	0.860	0.859
	(0.046)	(0.077)
Installer × city × year	0.853	0.848
	(0.050)	(0.081)
Difference (SMART − SCO), installer × year-month	−0.002 (0.087)	

Coefficients are pass-through (minus the coefficient on subsidy per kW). Each row replaces the installer-by-year fixed effects of column (4) of Table 4 with the stated fixed effects and keeps the other fixed effects and controls. The difference comes from one stacked regression with program-specific slopes, controls, and fixed effects, clustered by installer across programs. Discount rate 8%. Standard errors clustered by installer in parentheses.

B.4 Alternative Clustering

Table 19: Standard Errors under Alternative Clustering

Clustered by	Installer	City	Utility
Panel A: Solar Carve-Out			
Subsidy per kW	−0.863***	−0.863***	−0.863***
	(0.044)	(0.028)	(0.068)
Panel B: SMART			
Subsidy per kW	−0.847***	−0.847***	−0.847***
	(0.073)	(0.061)	(0.071)

Coefficient on subsidy per kW from the preferred specifications, column (4) of Table 4. The dependent variable is the net price per kW, and pass-through is minus the coefficient. Discount rate 8%. Standard errors in parentheses under each clustering. Utility clustering has only four clusters in the SMART sample and is shown for comparison only. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

B.5 Pooled versus Separate Estimation

Table 20 compares the separate estimates with pooled ones. Columns (1) and (2) come from one stacked regression in which the slope, the controls, and every fixed effect are specific to each program, clustered by installer across programs. Its estimates equal those from separate regressions, and the difference between the programs, reported below the two panels, has a standard error that accounts for installers active in both. Columns (3) to (6) impose one slope. With program-specific fixed effects (columns 3 and 4), the pooled slope is a weighted average of the two program slopes, with weights proportional to the residual variance of the subsidy in each program. Solar Carve-Out receives two-thirds to three-quarters of the weight because its sample is larger. With fixed effects shared across programs (columns 5 and 6), installer, equipment, and town effects are also assumed to be constant from 2014 to 2023.

Table 20: Pooled versus Separate Estimation

	Dependent variable: net price per kW					
	Slope by program		One slope, program FE		One slope, shared FE	
	Naive (1)	Cross-roof (2)	Naive (3)	Cross-roof (4)	Naive (5)	Cross-roof (6)
Panel A: Solar Carve-Out						
Subsidy per kW	−0.756***	−0.863***				
	(0.044)	(0.044)				
Panel B: SMART						
Subsidy per kW	−1.017***	−0.847***				
	(0.065)	(0.072)				
Difference in pass-through (SMART − SCO)	0.260***	−0.016				
	(0.090)	(0.085)				
Standard error if programs independent	(0.078)	(0.084)				
Subsidy per kW, both programs			−0.843***	−0.859***	−0.897***	−0.858***
			(0.031)	(0.037)	(0.038)	(0.037)
Weight on Solar Carve-Out			0.666	0.736		

Entries are coefficients on subsidy per kW, except the difference row, which reports the difference in pass-through. Pass-through is minus the coefficient. Naive: the specification of column (1) of Table 4. Cross-roof: column (4). Standard errors clustered by installer across programs in parentheses. The weight on Solar Carve-Out is its share of the residual variance of the subsidy after the fixed effects and controls. Discount rate 8%. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

Pless and van Benthem (2019) pool host-owned and third-party systems, which share one program, period, subsidy schedule, and set of fixed effects. The two programs here

differ in all four, so I let every coefficient differ by program. Four considerations support this choice. First, pooling imposes equal pass-through, which the naive specification rejects ($p = 0.004$). The pooled naive rate of 0.84 is then an average that describes neither program, and its weights reflect sample sizes rather than any policy question. Second, the paper's question is whether the two designs differ, which requires a slope for each. Third, the subsidies are measured differently. The SMART subsidy is a fixed contract, whereas the Solar Carve-Out subsidy depends on an assumed SREC price path, and pooling would let that assumption affect the SMART estimate. Fourth, the fixed effects must differ by program anyway. SMART needs utility-by-time effects because its rates differ by utility, the programs overlap only in 2018, and installer, town, and equipment effects plausibly changed between the two periods. Sharing them raises the naive pooled rate from 0.84 to 0.90, because it lets comparisons across programs, in which the subsidy fell by a factor of 3.4 while costs were falling, enter the estimate.

In the preferred specification, the two program estimates are nearly identical, and every version gives about 0.86. Pooling is then harmless, and 0.86 (0.04) is a fair summary of both programs.

B.6 Roof Tilt and Installation Cost

Table 21 reports the gross price per kW by five-degree tilt bin, relative to roofs pitched at 25–30 degrees, in the preferred specification with the subsidy per kW held fixed. Section 7.1 uses it.

Table 21: Gross Price per kW by Roof Tilt

	Panel A: Solar Carve-Out			Panel B: SMART		
Tilt (degrees)	$/kW	SE	Projects	$/kW	SE	Projects
0–5	372***	(122)	333	88	(67)	208
5–10	86**	(37)	407	25	(31)	322
10–15	6	(19)	773	−28	(24)	556
15–20	−19	(18)	2,598	13	(21)	1,556
20–25	−4	(16)	4,210	−6	(18)	2,431
30–35	42*	(21)	4,804	6	(17)	2,436
35–40	−12	(19)	3,518	−21	(17)	2,580
40–45	48*	(27)	2,045	17	(24)	1,292
45–50	−23	(55)	131	20	(69)	113
50–55	−165	(140)	16	137	(152)	18
55–60	162	(186)	11	−61	(193)	4
60+	199*	(118)	10	86	(251)	6
25–30 (reference)			4,647			2,795

Coefficients on tilt-bin indicators in a regression of the gross price per kW on the subsidy per kW, the tilt bins, and the controls and fixed effects of column (4) of Table 4. The reference bin is 25–30 degrees. Discount rate 8%. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

B.7 Pass-Through by Roof Productivity

This test estimates cross-roof pass-through by tercile of modeled capacity factor, using the specification in column (4) of Table 4, with terciles defined within each program. Roofs in the low tercile are the weak market and roofs in the high tercile the strong market. In one regression per program, with a separate slope on the subsidy and a separate intercept for each tercile and common fixed effects, pass-through is nearly the same in all three terciles: 0.98, 0.95, and 0.97 under Solar Carve-Out and 0.87, 0.83, and 0.87 under SMART. Equality

is not rejected (p = 0.66 and p = 0.46), and the difference between the high and the low tercile is −0.01 (0.04) in both programs. Estimating column (4) separately on each tercile's projects gives point estimates that fall from the low to the high tercile, from 0.95 to 0.81 under Solar Carve-Out and from 0.96 to 0.79 under SMART, but the standard errors are 0.07 to 0.20 and the differences are not significant. The estimates within terciles are higher than those in column (4) because the tercile intercepts remove the variation in generation between terciles. Measurement error in generation, which pulls the estimate toward one (Section 3.4), has a larger effect on the variation that remains. The comparison with installers' production estimates in Section 7.2 suggests that this effect can be large: within the cells of column (4), the two measures of the capacity factor have a correlation of only 0.57, and within terciles the true variation is smaller still. The data therefore do not show that pass-through differs between the weak and the strong market.

Table 22: Pass-Through by Tercile of Modeled Capacity Factor

	Dependent variable: net price per kW					
	One regression			Separate regressions		
	Low CF (1)	Middle CF (2)	High CF (3)	Low CF (4)	Middle CF (5)	High CF (6)
Panel A: Solar Carve-Out						
Subsidy per kW	−0.983***	−0.952***	−0.972***	−0.947***	−0.806***	−0.809***
	(0.069)	(0.059)	(0.058)	(0.075)	(0.162)	(0.198)
p-value, equal across terciles		0.658				
Panel B: SMART						
Subsidy per kW	−0.873***	−0.825***	−0.867***	−0.963***	−0.830***	−0.787***
	(0.091)	(0.095)	(0.086)	(0.140)	(0.147)	(0.160)
p-value, equal across terciles		0.459				

Coefficient on subsidy per kW by tercile of modeled capacity factor (CF), defined within each program, in the specification of column (4) of Table 4. Pass-through is minus the coefficient. Columns (1)–(3) come from one regression that interacts the subsidy with tercile indicators and includes tercile intercepts and common fixed effects, and the p-value is from a Wald test that the three slopes are equal. Columns (4)–(6) estimate column (4) of Table 4 separately on each tercile's projects. Discount rate 8%. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

B.8 Heterogeneity by Income and Installer Concentration

I examine whether SMART pass-through differs across ZIP codes by median household income and by local installer concentration. Income is the ZIP-level median household income from the American Community Survey (2016–2020 five-year estimates). Concentration is the Herfindahl–Hirschman index (HHI) of installers' shares of SMART projects in the ZIP code. I divide ZIP codes into terciles on each measure. For each measure, I estimate a single regression that interacts the subsidy with tercile indicators and keeps the fixed effects common across groups, and then test whether the three coefficients are equal. Table 23 reports the results.

Table 23: Heterogeneity in SMART Pass-Through

	Dependent variable: net price per kW							
	ZIP median income tercile				Installer HHI tercile			
Discount rate	3%	5%	8%	15%	3%	5%	8%	15%
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Subsidy per kW × tercile 1	−0.877***	−0.865***	−0.844***	−0.793***	−0.841***	−0.824***	−0.798***	−0.730***
	(0.058)	(0.064)	(0.073)	(0.098)	(0.067)	(0.074)	(0.085)	(0.114)
Subsidy per kW × tercile 2	−0.886***	−0.874***	−0.855***	−0.806***	−0.904***	−0.894***	−0.878***	−0.838***
	(0.063)	(0.070)	(0.080)	(0.107)	(0.059)	(0.065)	(0.074)	(0.099)
Subsidy per kW × tercile 3	−0.864***	−0.849***	−0.827***	−0.769***	−0.888***	−0.876***	−0.857***	−0.810***
	(0.068)	(0.075)	(0.086)	(0.115)	(0.059)	(0.065)	(0.075)	(0.100)
p-value, equal pass-through	0.845	0.845	0.845	0.845	0.216	0.216	0.216	0.216
Observations	14,195	14,195	14,195	14,195	14,195	14,195	14,195	14,195

SMART projects. Each column is one regression, with the subsidy per kW interacted with tercile indicators. Tercile 1 is the lowest-income tercile in columns (1)–(4) and the least concentrated tercile in columns (5)–(8). Entries are coefficients, so pass-through is minus each entry. All regressions use the preferred specification, column (4) of Table 4: utility-by-year-month, installer-by-year, city-by-year, inverter, and manufacturer fixed effects, log system size, and DC-to-AC ratio bins. Standard errors clustered by installer in parentheses. The p-value is from a joint Wald test that the three interaction coefficients are equal. It is the same at every discount rate because the discount rate rescales every SMART subsidy by the same factor (Appendix B.1). The sample has 14,195 projects, compared with 14,317 in the full SMART sample. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

Table 23 uses the preferred specification. Pass-through does not vary with income. At the 8% discount rate, the estimates are 0.84 in the lowest tercile, 0.85 in the middle tercile, and 0.83 in the highest tercile, and the hypothesis that they are equal is not rejected (p = 0.84). The pattern is mechanically the same at every discount rate (Appendix B.1). By concentration, the estimates are 0.80, 0.88, and 0.86 from the least to the most concentrated

tercile, so pass-through is lowest where installers are least concentrated, the opposite of what market power would predict, and the equality test does not reject ($p = 0.22$). In the naive specification, pass-through falls with income, from 1.09 to 0.91 ($p = 0.011$), but this gradient disappears once the timing of the subsidy is absorbed.

Two cautions apply. First, the pattern is not robust to how the fixed effects are estimated. If I estimate the model separately within each tercile, allowing every fixed effect to differ by group, the estimates are 0.74, 0.82, and 1.00 from the lowest to the highest income tercile, and 0.85, 0.94, and 0.79 across concentration terciles, with no monotone pattern by concentration (Figures 8 and 9). The split samples are a third of the full sample with installer-by-year and city-by-year effects estimated within each, so these estimates are imprecise, and I do not put weight on either set of gradients. Second, income and concentration are measured at the ZIP level from SMART records only. Income is an ACS estimate for the ZIP, not the buyer's own income, and the HHI is computed from the same projects used in the regression, so ZIPs with few projects have mechanically high concentration. Terciles are equal-sized groups of projects and not of ZIP codes.

Taken together, the results give no evidence that pass-through differs with local income or that it is lower where installers are more concentrated. The second finding bears on the interpretation of the main results: if pass-through under SMART reflected market power, one would expect it to vary with local competition and, since demand is log-concave (Section D.1), to be lower where competition is weaker. The absence of this gradient is consistent with the view that market power is not the main determinant of pass-through in this market, but the test has limited power, since the HHI measure varies within a range of low concentration statewide (Appendix C.3.3).

Figures 8 and 9 report SMART pass-through when the model is estimated separately within income and concentration terciles, so that every fixed effect can differ by group.

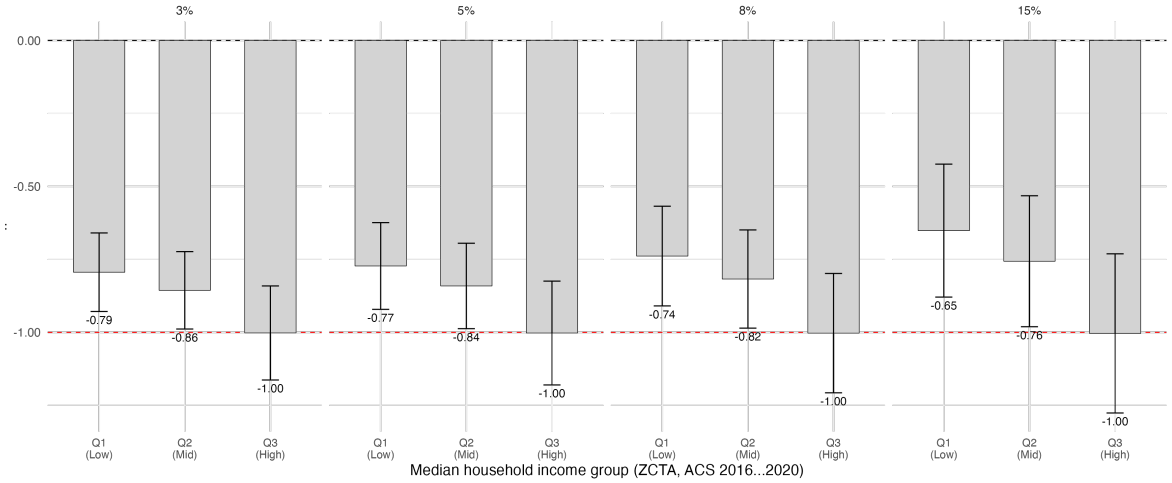


Figure 8: SMART Pass-Through by ZIP Income Tercile, Separate Regressions

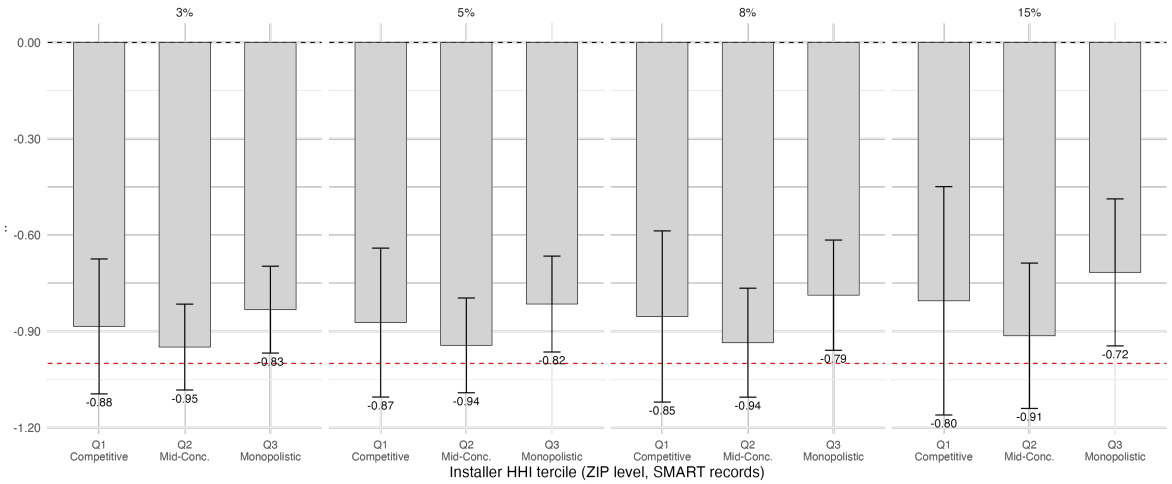


Figure 9: SMART Pass-Through by Installer Concentration Tercile, Separate Regressions

C Timing-Based Designs: Additional Results

C.1 Naive Specifications

Table 24 reports the naive model as controls are added. Columns (1) and (5) control only for year fixed effects in Panel A (Solar Carve-Out) and for utility and year fixed effects in Panel B (SMART). Columns (2) and (6) add inverter and panel manufacturer fixed effects. Columns (3) and (7) add installer and city fixed effects. Columns (4) and (8) include all controls and are column (1) of Table 4.

Table 24: Naive Pass-Through Estimates: Solar Carve-Out and SMART

	Panel A: Solar Carve-Out				Panel B: SMART			
	Dependent Variable: Net Price per kW				Dependent Variable: Net Price per kW			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
Subsidy per kW	−0.709***	−0.729***	−0.751***	−0.756***	−0.908***	−0.888***	−1.005***	−1.017***
	(0.109)	(0.073)	(0.042)	(0.044)	(0.138)	(0.089)	(0.061)	(0.065)
Controls								
Inverter		✓		✓		✓		✓
Manufacturer		✓		✓		✓		✓
Installer			✓	✓			✓	✓
City			✓	✓			✓	✓
Utility FE					✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Observations	25,491	25,491	25,491	25,491	14,317	14,317	14,317	14,317

Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1. Subsidies are discounted at 8% with 0.5% annual panel degradation. SREC subsidies use the midpoint of the ACP and auction prices.

Across specifications in Panel A, the naive estimates with few or many controls are all close to the full model, so the estimate is not driven by variation in the cost factors that the controls capture. Specifications in Panel B are less similar to one another. Adding manufacturer and inverter controls alone lowers the estimate slightly (0.888), while adding installer and city fixed effects raises it to 1.005, very close to the full model (1.017). The standard errors fall as controls are added, from 0.138 to 0.065 in Panel B, which is consistent with the controls removing cost variation that is unrelated to the subsidy.

C.2 Price Level

The naive specification is in nominal dollars, for the reasons given in Section 4.1. The SMART estimate is, however, sensitive to how price-level changes are treated: it is 1.02 in nominal dollars, 0.91 when the CPI is added as a control, 0.93 with year-by-month fixed effects, and 0.85 in CPI-deflated dollars (Table 25), while the Solar Carve-Out estimate stays between 0.73 and 0.80. Consumer prices rose by about 22% over the SMART sample while rates were stepping down, so each of these alternatives also removes part of the co-movement between the subsidy and costs discussed in Section 6.1. Because the five specifications use the same projects, I test whether they differ with one stacked regression that gives each specification its own slope, controls, and fixed effects and clusters standard errors by installer across the stacked copies. The SMART estimates differ significantly: every alternative is below the naive estimate, by 0.11 (standard error 0.03) with the CPI control, 0.09 (0.04) with year-by-month fixed effects, 0.17 (0.01) in CPI-deflated dollars, and 0.13 (0.05) in CPI-deflated dollars with year-by-month fixed effects, and equality of all five is rejected ($p < 0.001$). Under Solar Carve-Out the differences are small, from −0.03 to 0.04, but precisely estimated, and equality is also rejected. That the SMART estimate falls whenever part of the rise in the price level is removed is consistent with the naive estimate being pushed up by the timing of the subsidy.

Table 25: Pass-Through under Alternative Treatments of the Price Level

	Dependent variable: net price per kW				
	Nominal dollars			CPI-deflated dollars	
	Year FE (naive) (1)	Year FE, CPI control (2)	Year-by-month FE (3)	Year FE (4)	Year-by-month FE (5)
Panel A: Solar Carve-Out					
Subsidy per kW	−0.756***	−0.760***	−0.799***	−0.729***	−0.797***
	(0.044)	(0.046)	(0.046)	(0.042)	(0.045)
Panel B: SMART					
Subsidy per kW	−1.017***	−0.906***	−0.929***	−0.846***	−0.883***
	(0.065)	(0.063)	(0.062)	(0.062)	(0.062)

Full specifications with installer, inverter, manufacturer, and city fixed effects (and utility fixed effects for SMART). Pass-through is minus the coefficient on subsidy per kW. Discount rate 8%. Deflated values use the monthly CPI-U and are expressed in dollars of the first half of 2019 (index 254.412). The CPI ranges from 234 to 253 over the Solar Carve-Out sample and from 248 to 302 over the SMART sample. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

C.3 Other Explanations for the Naive Gap

The naive estimates differ between the programs: 0.76 for Solar Carve-Out and about one for SMART. Section 6 shows that most of this gap disappears once the timing of the subsidy is absorbed, so the main explanation is how each subsidy moved over time relative to installation costs. This section examines three further candidate explanations. None of the three can be isolated cleanly, because the programs differ in size, certainty, period, and eligible population at the same time, so the evidence here is descriptive. The equal cross-roof estimates suggest that none of them is large.

C.3.1 Size

The first explanation is that SMART subsidies are small enough to be absorbed without a supply response, whereas Solar Carve-Out subsidies, worth about 45% of the gross price, strain installers' capacity. The mean subsidy is $1,923 per kW under Solar Carve-Out and $566 per kW under SMART, and Section 3 shows that the supports of the two distributions barely overlap. A direct comparison at the same subsidy level is therefore not possible. What I can do is ask whether pass-through falls with the size of the subsidy within each program. Table 26 reports naive pass-through by quartile of subsidy per kW. Under Solar Carve-Out the estimate is flat, between 0.77 and 0.78 in every quartile, so within that program larger subsidies are not passed through less. Under SMART the estimate falls from 1.55 in the lowest quartile to 1.09 in the highest, which is the direction the size explanation predicts. However, the lowest SMART quartile has a median subsidy of $127 per kW and includes projects with no subsidy at all, so the estimate is imprecise (a standard error of 0.23), and the gradient may also reflect rising costs in the later SMART years, when the subsidy was smallest (Figure 4). Estimates above one in the two lowest quartiles are hard to reconcile with log-concave demand (Section 6.5), which favors the second reading. Taken together, the within-program evidence is mixed: it does not support the size explanation under Solar Carve-Out and is consistent with it, but confounded, under SMART. The workload test in

Section 8.4 points the same way for Solar Carve-Out: if large subsidies strained installers' capacity, pass-through would be lower when installers are busy, and it is not.

Table 26: Pass-through by Quartile of Subsidy per kW

	Q1	Q2	Q3	Q4
Panel A: Solar Carve-Out				
Median subsidy ($/kW)	1,528	1,698	2,151	2,305
Subsidy per kW	−0.779***	−0.776***	−0.780***	−0.770***
	(0.073)	(0.068)	(0.062)	(0.059)
Panel B: SMART				
Median subsidy ($/kW)	127	458	669	947
Subsidy per kW	−1.552***	−1.317***	−1.189***	−1.091***
	(0.228)	(0.131)	(0.091)	(0.076)

Coefficient on subsidy per kW in each quartile, from a regression of the net price per kW in which subsidy per kW is interacted with quartile indicators of the same variable, with the full set of fixed effects. Pass-through is minus the coefficient. Discount rate 8%. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

C.3.2 Predictability

The second explanation is that fixed SMART rates reduce uncertainty. Buyers who cannot forecast SREC revenue may value it less than its expected present value, and installers cannot raise prices as far. Three observations bear on this. First, the ACP capped the SREC price and the auction price was meant to support it, so the range of likely prices was narrower than in an unregulated market. Market prices nonetheless fell below the auction price for several years (Figure 1), so the auction price did not act as a floor in practice. Second, the estimate is not sensitive to how one forecasts prices: in the naive specification it ranges from 0.723 (auction price) to 0.782 (ACP) across the three assumptions and is 0.747 with actual market prices, and in the preferred specification it ranges from 0.844 to 0.878 (Table 14). If uncertainty about the price path were the main driver, the estimate should be more sensitive to the assumed path. Third, naive pass-through under Solar Carve-Out falls with the discount rate (0.805 at 3% versus 0.682 at 15%, and 0.890 versus 0.821 in the preferred specification), whereas the SMART estimate cannot respond to

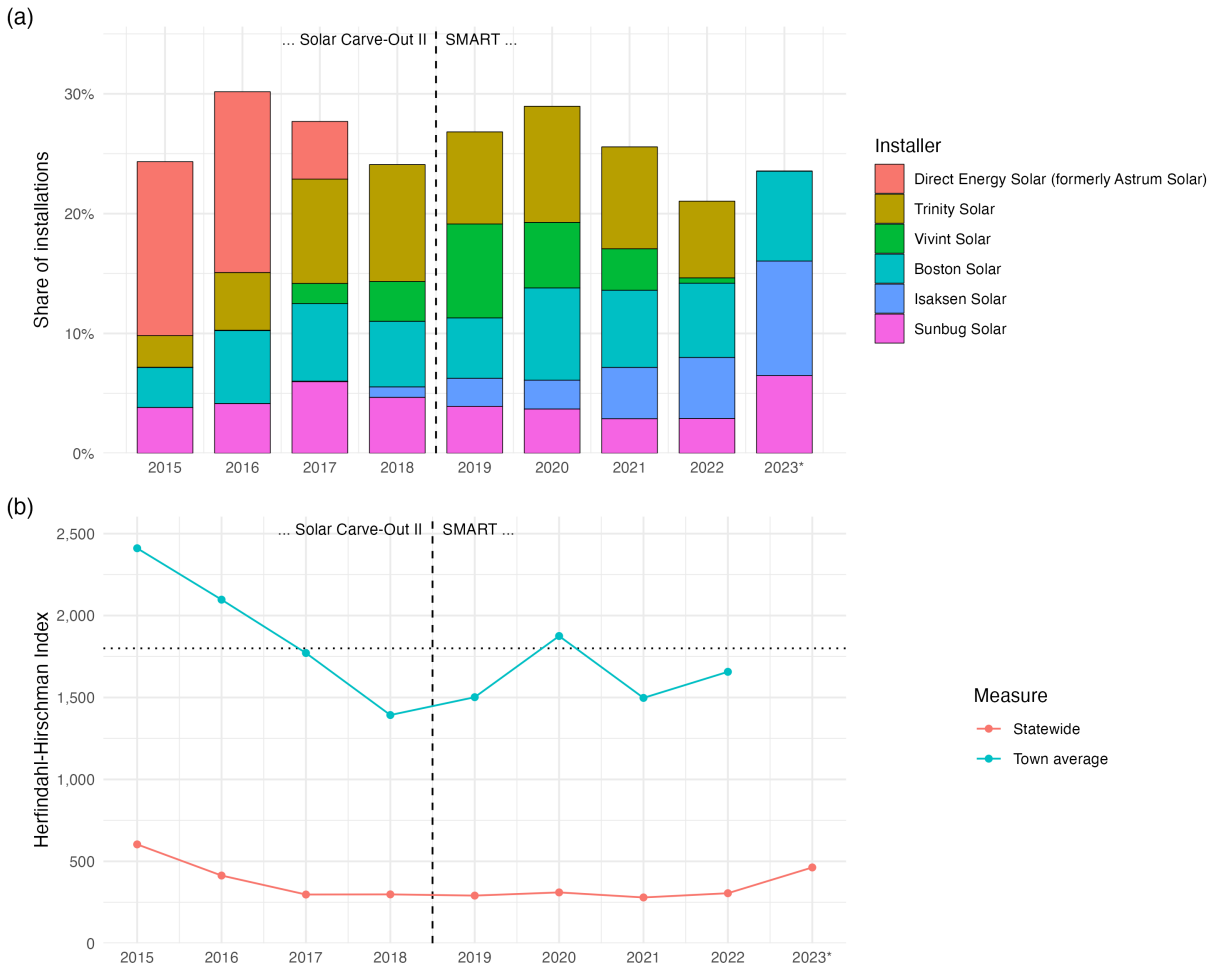


Figure 10: Installer Market Share and Concentration

it by construction (Appendix B.1). The Solar Carve-Out pattern is consistent with buyers applying high discount rates to uncertain, back-loaded certificate revenue, but it is also partly the mechanical effect of a larger regressor.

C.3.3 Market Competition

The third explanation concerns competition. Figure 10 shows the concentration of installers over the sample. Panel (a) shows the market share of the four largest installers in each program, and panel (b) shows the Herfindahl–Hirschman index (HHI) at the statewide and town levels.

At the state level, the market is unconcentrated throughout. The statewide HHI is 603

in 2015 and falls to about 300 by 2017, where it stays between 280 and 310 through 2022. The top four installers' combined share falls from about 40% in 2015 to 23–27% from 2017 on. The number of active installers per year is 184–216 in 2015–2018 and 128–162 in 2019–2022. The figure starts in 2015 because Solar Carve-Out II opened in April 2014 and, in that partial year, only 28 towns have at least 20 installations, so the 2014 town average (above 6,000) is not comparable with later years. Town-level concentration, which is more relevant for a household choosing among installers who quote locally, is higher: the project-weighted town HHI falls from about 2,400 in 2015 to about 1,400 in 2018 and is 1,500–1,900 in the SMART years with no downward trend within the period.

These patterns do not support the idea that the market became less competitive during SMART, nor do they show that competition intensified in a way that could explain the jump in pass-through. Most of the fall in concentration happened before SMART started, between 2015 and 2017, when Solar Carve-Out pass-through was already about 0.76. Within each program, pass-through rises only slightly with local concentration, and the change is not significant at the 5% level (Section 8.4 and Appendix B.8). Together with low statewide concentration, this suggests that market power is not the main reason for the gap between the two programs.

D Demand Estimation

D.1 Main Estimates

The bound in Section 8.3 requires the curvature of demand. I estimate demand following Pless and van Benthem (2019). The unit is a ZIP code and year, and the outcome is installations per 1,000 residents. Demand is quadratic in the mean net price per watt,

$$q_{zt} = \beta_1 p_{zt} + \beta_2 p_{zt}^2 + X_z \gamma + \delta_{c(z)} + \tau_t + e_{zt}, \tag{3}$$

so that β_2 measures convexity. The vector X_z contains ZIP characteristics from the 2017–2021 American Community Survey (log median household income, the share of adults with a bachelor's degree, median age, log median home value, and the share of owner-occupied homes), $\delta_{c(z)}$ are county fixed effects, and τ_t are year fixed effects. The net price responds to demand shocks, so I instrument p and p^2 with the mean subsidy per watt and its square and with the average weekly wage in the county's building-equipment contracting sector (NAICS 2382, from the Quarterly Census of Employment and Wages) and its square. The subsidy shifts the price buyers pay, and the wage shifts installation costs. Neither should affect adoption except through price. This is the functional form that Pless and van Benthem (2019) use, with the same types of instruments, so a difference in curvature between California and Massachusetts is not an artifact of the specification. Standard functional forms impose the curvature of demand: linear demand is log-concave, exponential demand is log-linear, and constant-elasticity demand is log-convex (Fabinger and Weyl, 2014). I use a quadratic because it is the lowest-order form that estimates curvature rather than imposing it, and I evaluate curvature at the mean price because demand curves need not have constant curvature. Log-concavity is also what one would expect for a discrete adoption decision when willingness to pay has a log-concave density (Bagnoli and Bergstrom, 2005). Appendix D.3 replaces the quadratic with a natural spline. From the estimates, I compute

the price elasticity and the curvature $1/\varepsilon_{ms} = 1 - qq''/q'^2$ at the mean price, with standard errors by the delta method. Because the quadratic in levels imposes a particular shape, I also estimate the same equation with $\log q_{zt}$ as the outcome. In that form the coefficient on p^2 is zero exactly when demand is log-linear, so it is a direct test of log-linear demand, and $1/\varepsilon_{ms} = -2\beta_2/(\beta_1 + 2\beta_2 p)^2$. The sample includes ZIP-years with at least one installation and at least 500 residents.

Table 27: Demand Estimates

	Panel A: Solar Carve-Out		Panel B: SMART	
Dependent variable	Quadratic q (1)	Log-quadratic $\log q$ (2)	Quadratic q (3)	Log-quadratic $\log q$ (4)
Net price	0.777	2.110	2.516***	2.343***
	(4.257)	(2.690)	(0.605)	(0.563)
Net price squared	−0.124	−0.362	−0.439***	−0.446***
	(0.716)	(0.459)	(0.082)	(0.075)
Price elasticity at the mean	0.27	0.79	−1.57***	−2.16***
	(1.21)	(1.09)	(0.42)	(0.42)
Curvature $1/\varepsilon_{ms}$ at the mean	14.86	7.05	5.38*	2.14**
	(56.01)	(11.35)	(2.77)	(1.02)
First-stage Wald statistic, p / p^2	20.3 / 16.3	20.3 / 16.3	61.2 / 56.3	61.2 / 56.3
Mean net price ($/W)	2.47	2.47	3.35	3.35
Mean installations per 1,000 residents	1.54	1.54	0.91	0.91
ZIP-years	2,171	2,171	1,715	1,715

Two-stage least squares estimates. Columns (1) and (3): equation (3), with q the installations per 1,000 residents. Columns (2) and (4): the same equation with $\log q$ as the outcome, so that the coefficient on the squared price is zero under log-linear demand. Instruments: mean subsidy per watt and its square, and the county building-equipment contractor wage and its square. Controls: ACS 2017–2021 ZIP characteristics, county and year fixed effects. $1/\varepsilon_{ms} = 0$ for log-linear demand, > 0 for log-concave demand, and < 0 for log-convex demand. Standard errors clustered by ZIP code in parentheses. Elasticity and curvature standard errors by the delta method. Appendix D.2 reports alternative specifications. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

Table 27 reports the estimates. Under SMART, demand slopes down with an elasticity of −1.57 at the mean price of $3.35 per watt. This is nearly twice the −0.85 that Pless and van Benthem (2019) report for host-owned systems in California at a mean price of $3.70 per watt.[27] The squared term is negative and precisely estimated, so demand is concave, and the

[27]The difference may reflect a later period in which buyers had more alternatives to buying a system outright, such as leases and power purchase agreements, which are outside my sample. It may also reflect

curvature measure is 5.38. The log-quadratic form gives the same answer more sharply: the coefficient on the squared price is −0.446, which rejects log-linear demand, and the curvature is 2.14. Demand under SMART is therefore log-concave, not log-linear and not log-convex, whereas Pless and van Benthem (2019) find convex demand in California with the same functional form.[28] The result holds across specifications. The elasticity ranges from −1.14 to −1.92 and the curvature from 3.4 to 9.8 when I add utility fixed effects, replace county and year fixed effects with county-by-year fixed effects, weight by population, drop ZIP-years with fewer than three installations, or cluster by county (Appendix D.2). Under Solar Carve-Out, demand is not identified. The subsidy is driven by the statewide certificate price and varies little across ZIP codes within a county and year, and the estimates change sign across specifications. I therefore draw structural conclusions only for SMART.

Two further limitations qualify the SMART result. First, the curvature is measured with limited precision: the quadratic curvature of 5.38 has a t-statistic of about 1.9 and the log-quadratic curvature of 2.14 about 2.1, and the bound on market power below depends on its sign. Second, the sample excludes ZIP-years with no installations, which truncates demand at low quantities, and truncation affects the estimated curvature directly.

who is buying: early adopters are more likely to install at almost any price, so later buyers may simply be more price-responsive. Differences in samples and instruments may also contribute.

[28] Both forms bend back at low prices: the level quadratic slopes upward below \$2.87 per watt (a quarter of ZIP-years) and the log-quadratic below \$2.62 (13%), so the estimates describe demand around the mean rather than over the whole range.

D.2 Alternative Specifications

Table 28 reports the quadratic demand estimates of Section D.1 under alternative specifications. With county-by-year fixed effects and only the subsidy instruments, the log-quadratic SMART estimates are almost unchanged: the coefficient on the squared price is −0.425 (0.103) and the curvature is 2.13 (1.12). Under SMART, the price coefficients and the curvature keep their sign in every specification. Under Solar Carve-Out, the estimates are imprecise and change sign, so demand is not identified. I also replaced the subsidy instruments with the mean modeled capacity factor of each ZIP code's projects and its square, which shift the subsidy only through roof geometry. With the wage instruments, the first-stage Wald statistics are 7.6 and 8.6, and the curvature is positive but has standard errors of 13 (log-quadratic) and more than 1,000 (quadratic). Without them, the first stage has no power. These instruments are too weak to test the subsidy instrument.

Table 28: Demand Estimates under Alternative Specifications

	Baseline	Utility FE added	County × year FE, subsidy IV	Population-weighted	ZIP-years with 3+ installations	Clustered by county
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Solar Carve-Out						
Price elasticity at the mean	0.27	0.30	−2.51	2.24	−4.93**	0.27
	(1.21)	(1.29)	(1.61)	(1.40)	(2.46)	(1.12)
Net price squared (β_2)	−0.124	−0.216	−0.202	−1.001	5.026**	−0.124
	(0.716)	(0.765)	(0.909)	(0.857)	(2.132)	(0.768)
Curvature $1/\varepsilon_{ms}$ at the mean	14.86	19.58	1.25	2.59***	−0.42	14.86
	(56.01)	(97.34)	(1.43)	(0.78)	(0.91)	(58.00)
ZIP-years	2,171	2,171	2,171	2,171	1,809	2,171
Panel B: SMART						
Price elasticity at the mean	−1.57***	−1.92**	−1.62***	−1.21***	−1.14**	−1.57***
	(0.42)	(0.77)	(0.43)	(0.37)	(0.47)	(0.51)
Net price squared (β_2)	−0.439***	−0.417***	−0.251***	−0.216***	−0.570***	−0.439***
	(0.082)	(0.092)	(0.082)	(0.062)	(0.116)	(0.122)
Curvature $1/\varepsilon_{ms}$ at the mean	5.38*	3.80	3.35*	4.64	9.78	5.38
	(2.77)	(2.65)	(1.73)	(2.97)	(8.39)	(3.55)
ZIP-years	1,715	1,715	1,715	1,715	1,237	1,715

Two-stage least squares estimates of equation (3). Elasticity and curvature at the mean net price and installation rate of each sample. Baseline: county and year fixed effects, with the subsidy per watt, the county building-equipment contractor wage, and their squares as instruments. Column (3) uses only the subsidy instruments, because the wage varies at the county-year level. Standard errors clustered by ZIP code (by county in column 6) in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

D.3 Flexible Functional Form

The quadratic in Section D.1 estimates curvature from the data but restricts how curvature varies with price. To check whether this restriction matters, I replace the quadratic with a natural cubic spline in the net price, with three to five degrees of freedom and interior knots at quantiles of the price distribution. A spline enters the endogenous price through several nonlinear terms, and two-stage least squares with polynomials of the instruments identifies them only weakly: with nine instruments, the curvature at the mean ranges from −87 to 1.9 across three to five degrees of freedom, with standard errors of up to 389. I therefore use a control function (Newey, Powell and Vella, 1999). The first stage regresses the price on the instruments, controls, and fixed effects of equation (3), and the demand equation adds the first-stage residual to the spline. Standard errors come from 499 bootstrap draws that resample ZIP codes and re-estimate both stages. Under Solar Carve-Out, the curvature estimates have standard errors in the thousands under every form, consistent with demand not being identified (Section D.1), so Table 29 reports SMART only.

Three findings stand out. First, curvature is positive in every specification, between 0.73 and 1.51 at the mean price, and 97 to 100 percent of bootstrap draws imply log-concave demand. The conclusion that demand under SMART is log-concave therefore does not depend on the quadratic. Second, with the estimator held fixed, the quadratic and the splines give similar curvature (1.31 for the quadratic and 0.73 to 1.51 for the splines), so the quadratic is flexible enough near the mean. Third, the baseline curvature of 5.38 is larger than under any flexible form. Figure 11 shows why. The baseline quadratic turns upward at \$2.87 per watt, not far below the mean of \$3.35, and the curvature of a quadratic rises steeply as its slope approaches zero. From about \$3.80 per watt upward, all three curves in the figure lie between about one and two.

Smaller curvature loosens the bound. At the cross-roof pass-through estimate with modeled generation, the implied markup is at most 4.1 to 7.0 percent of price under the control-function estimates, compared with 2.1 percent for the baseline quadratic and 3.9 percent

for the log-quadratic form. With the sensitivity values of pass-through from the installer's estimate, 0.77 and 0.74, the control-function bounds are 6.6–11.3 and 8.0–13.7 percent (Table 11). The baseline quadratic gives the tightest bound.

Table 29: SMART Demand Curvature under Flexible Functional Forms

	Two-stage least squares		Control function			
	Quadratic (baseline)	Log-quadratic	Quadratic	Natural spline		
				3 df	4 df	5 df
	(1)	(2)	(3)	(4)	(5)	(6)
Elasticity at the mean	−1.57***	−2.16***	−2.54***	−3.01***	−2.94***	−3.52***
	(0.42)	(0.42)	(0.36)	(0.41)	(0.39)	(0.76)
Curvature $1/\varepsilon_{ms}$ at the mean	5.38*	2.14**	1.31***	1.32***	1.51***	0.73
	(2.77)	(1.02)	(0.17)	(0.14)	(0.56)	(0.98)
Share of bootstrap draws log-concave			1.00	1.00	1.00	0.97
Markup bound (% of price)	2.1	3.9	5.5	4.6	4.1	7.0

SMART ZIP-years (1,715). Elasticity and curvature at the mean net price and installation rate. Columns (1) and (2): two-stage least squares from Table 27, with standard errors clustered by ZIP code. Columns (3)–(6): control function, in which the demand equation includes the first-stage residual of the price. Natural splines have interior knots at quantiles of the net price. Standard errors in columns (3)–(6) and the share of log-concave draws come from 499 bootstrap draws that resample ZIP codes and re-estimate both stages. The markup bound is $(1/\rho_B - 1)/(1/\varepsilon_{ms})$ divided by the absolute elasticity, at $\rho_B = 0.847$ (Table 4, column 4). All specifications include the ZIP controls and county and year fixed effects of equation (3). Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

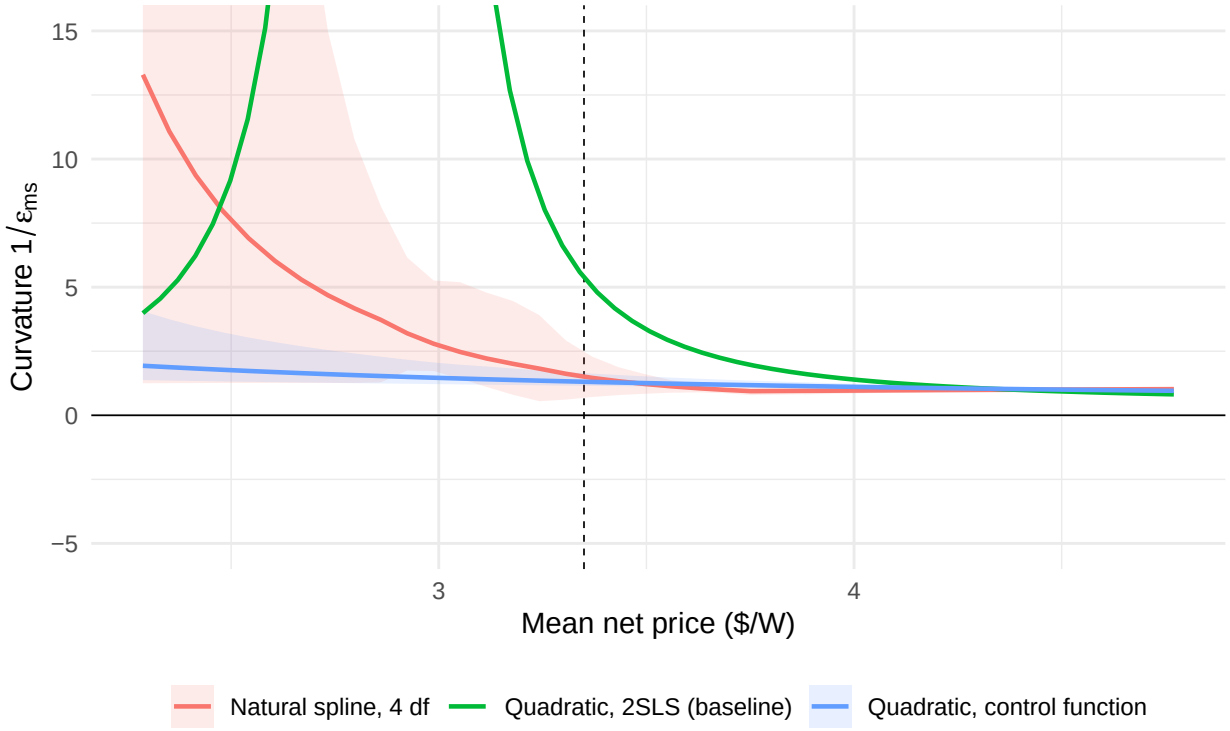


Figure 11: Curvature of SMART Demand across Prices. Curvature $1/\varepsilon_{ms}$ evaluated along each fitted demand curve between the 5th and 95th percentiles of the net price. Shaded areas are 90% bootstrap intervals for the control-function estimates. The dashed line marks the mean net price.

E Pass-Through of a Roof-Specific Subsidy

This appendix derives the expression for roof-specific pass-through in Section 8.1. Group buyers by roof type k, which determines the subsidy s_k a buyer receives, and let $D_k(p)$ be the probability that a buyer of type k installs at net price $p = P_k - s_k$, where P_k is the gross price the installer charges. The installer observes k, because it computes the roof's expected output itself, and its cost of installing roof k is c_k, which does not depend on the subsidy. Following Weyl and Fabinger (2013), I summarize conduct by a parameter $\theta_B \in [0, 1]$ in the pricing condition

$$P_k - c_k = \theta_B \, m_k(P_k - s_k), \qquad m_k(p) \equiv -\frac{D_k(p)}{D_k'(p)},$$

which is marginal-cost pricing when $\theta_B = 0$ and monopoly pricing of roof k when $\theta_B = 1$. Intermediate values describe competition among the installers that quote the roof. Because one roof is a negligible share of an installer's volume, a change in s_k does not move the installer along its marginal cost curve, so c_k is fixed.

Differentiating with respect to s_k,

$$\frac{dP_k}{ds_k} = \theta_B \, m_k'(p) \left(\frac{dP_k}{ds_k} - 1 \right) \quad \Longrightarrow \quad \frac{dP_k}{ds_k} = \frac{-\theta_B m_k'}{1 - \theta_B m_k'}.$$

Since $m_k' = -(1 - D_k D_k''/D_k'^2) = -1/\varepsilon^B_{ms}$, the pass-through of the roof's subsidy to the buyer's net price is

$$\rho_B = 1 - \frac{dP_k}{ds_k} = \frac{1}{1 + \theta_B/\varepsilon^B_{ms}}, \qquad \text{so} \qquad \frac{1}{\rho_B} - 1 = \frac{\theta_B}{\varepsilon^B_{ms}}.$$

For a subsidy that changes for all buyers at once, the installer's total volume changes, and the slope of marginal cost adds the term r of equation (2). With $r \geq 0$ and the same conduct and curvature in the two expressions, the market-wide rate is therefore no higher than the roof-specific rate.

Three cases give $\rho_B = 1$ regardless of the installer's markup. Under perfect competition, $\theta_B = 0$. If the installer charges one price per kW for every roof, P_k cannot depend on s_k. And

if demand is log-linear, $1/\varepsilon^B_{ms} = 0$, and the markup m_k is a constant number of dollars. Pass-through below one therefore requires both pricing by roof and log-concave demand among buyers with similar roofs.

Two qualifications apply to the bound. First, if installation cost varies with the subsidy across roofs, $c_k = c + \kappa s_k$, then dP_k/ds_k gains the term $\kappa/(1 - \theta_B m'_k)$, and the estimated shortfall $1 - \hat{\rho}_B$ overstates the shortfall due to conduct when $\kappa > 0$ and understates it when $\kappa < 0$. Section 7.1 shows that, holding tilt fixed, cost differences across roofs leave the SMART estimate unchanged. Second, the bound uses the curvature of aggregate ZIP-level demand in place of ε^B_{ms}, the curvature among buyers with similar roofs, which the data do not identify separately.

F Price Discrimination Across Roofs

Installers quote each project individually and compute the roof's expected output themselves, as the production estimates on SMART applications show, so the installer knows the subsidy a buyer will receive. Section 8.1 shows that the cross-roof estimates measure the pass-through of a roof-specific subsidy, which is below one only if the gross price rises with the roof's subsidy, and Table 10 shows that installers price the roof only a little. Under SMART the rate is also known with certainty at the time of sale. If certainty made pricing by roof easier, cross-roof pass-through would be lower under SMART than under Solar Carve-Out. The two estimates are nearly identical (0.85 and 0.86), which gives no indication that it did.

F.1 Welfare Effects of Pricing by Roof

If installers do price by roof, a natural question is whether buyers as a group gain or lose relative to a uniform price. Aguirre, Cowan and Vickers (2010) and Miklós-Thal and Shaffer (2021) show that the answer depends on pass-through and on the curvature of demand in each market. Treat roofs with high generation as the strong market, where the discriminatory price is higher, and roofs with low generation as the weak market. Relative to uniform pricing, discrimination raises output if pass-through is higher in the weak market ($\rho_W > \rho_S$, averaged over the relevant range) and inverse demand is convex in both markets. It lowers output if $\rho_W < \rho_S$ and inverse demand is concave in both markets. The same conditions, with pass-through weighted by the markup, determine the welfare effect. Miklós-Thal and Shaffer (2021) show that the ranking of pass-through rates alone is not sufficient, so the sign of curvature is needed. Demand under SMART is concave (Table 27). If pass-through were lower on roofs with low generation, pricing by roof would therefore reduce adoption relative to a uniform price per kW. The test in Appendix B.7 finds no such difference, so the data do not sign this effect. Miklós-Thal and Shaffer (2021) also note that formulas based on

pass-through hold exactly only for small changes unless demand is linear, so the division of surplus in Section 8.3, which is evaluated at the mean, is an approximation for a subsidy of this size.